

The Handbook of Basic Sericulture has been prepared to serve the special purpose of college students. This book explain the Basic Sericulture Techniques in a simplified manner. This book can be used for UG and PG Zoology students,who are looking for a quick revision before the Examination. This book support the Certificate and Diploma sericulture studying students to get confidence and also reference just before attending Department of Sericulture Competitive Examination. This book has been written to students and farmers the field of Sericulture ,No such book which cover suitable original picture with needed explanation of Sericulture Techniques and Practice. This book is useful for the preparation of State Universities and College Students integrated Certificate and Diploma courses of sericulture.



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Kalaimani Subramani

The Handbook of Basic Sericulture

(Nut Cell of Basic Sericulture Rearing Techniques)

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“THE HAND BOOK OF BASIC SERICULTURE”



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“THE HAND BOOK OF BASIC SERICULTURE”

CONTENT

UNIT	TITLE	PAGE NO
1.	LIFE CYCLE OF SILK WORM	5
2.	TYPES OF SILK WORM	7
3.	VOLTINISM	11
4.	MULBERRY CULTIVATION	13
5.	METHODS OF HARVESTING	28
6.	PRESERVATION OF LEAVES	30
7.	PEST OF MULBERRY	32
8.	SILK WORM REELING	36
9.	METHODS OF STIFLING	38
10.	SILK REELING	41
11.	RE REELING	44

12.	LACING,SKEINING	46
13.	BOOK MAKING	47
14.	EXAMINATION OF SILK	48
15.	CLASSIFICATION OF COCOON	50
16.	COCOON TESTING AND GRADIND	52
17.	SILKWORM REARING	58
18.	SILKWORM REARING TECHNIQUE	64
19.	REFERENCE	87

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S. KALAIMANI

THE HAND BOOK OF BASIC SERICULTURE

LIFE CYCLE OF SILKWORM

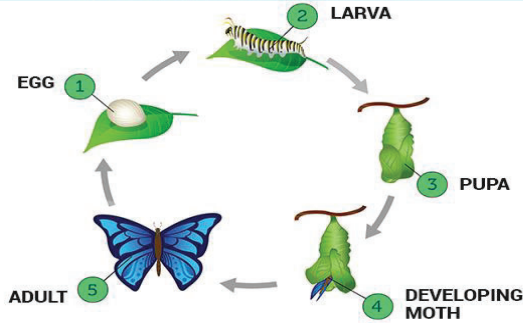
The life cycle of silk moth starts when a female silk moth lays eggs. The caterpillar or larvae are hatched from the eggs of the silk moth. The silkworms feed on mulberry leaves and give rise to pupa.

In pupa stage, a weave is netted around by the silkworm to hold itself. After that it swings its head, spinning a fiber made of a protein and becomes a silk fiber. Several caterpillars form a protective layer around pupa and this covering is known as the cocoon.

The silk thread (yarn) is obtained from the silk moth's cocoon. The life cycle of silkworm is explained below in details.



LIFE CYCLE OF SILKWORM



Stage 1: EGG

An egg is the first stage of the life cycle of silkworm. The egg is laid by a female moth which is mostly the size of small dots. A female moth lays more than 350 eggs at a time. In the springtime, the eggs hatch due to the warmth in the air. This procedure happens once in every year.

Stage 2: SILKWORM

A hairy silkworm arises after the eggs crack. In this stage of silkworms, the growth happens. They feed on mulberry leaves and consume a large amount of these leaves for around 30 days before going to the next stage.

Stage 3: COCOON

In this stage, silkworms spin a protective cocoon around itself. It is the size of a small cotton ball and is made of a single thread of silk.

Stage 4: PUPA

Pupa stage is a motionless stage. In this stage, people kill the pupa by plunging the cocoon into boiling water and unwind the silk thread.

Stage 5: MOTH

In this stage, the pupa changes into an adult moth. The female moth lays eggs after mating and thus the life cycle of silkworm begins again.

TYPES OF SILK WORMS

1. MULBERRY SILKWORM

- The bulk of the commercial silk produced in the world comes from this variety and often silk generally refers to mulberry silk.
- Mulberry silk comes from the silkworm, *Bombyx mori* L. which solely feeds on the leaves of mulberry plant.
- These silkworms are completely domesticated and reared indoors. In India, the major mulberry silk producing states are Karnataka, Andhra Pradesh, West Bengal, Tamil Nadu and Jammu & Kashmir which together accounts for 92 % of country's total mulberry raw silk production.
- Silk insects other than mulberry silk insects are called non-mulberry silk insects. As they are reared outdoors in wild, they are also known as wild silk insects.
- The only exception is the domesticated eri silk moth, *Samia ricini*.
- All the commercially exploited wild silk moths of India belong to the family, Saturniidae which are capable of flight and follow the same pattern of life cycle.

2. MUGA SILKWORM:

- Muga silkworm (*Antheraea assamensis*) produces silk of golden yellow lustre. The popular name 'Muga' is an Assamese word meaning brown colour.
- Muga silk is grown exclusively in the North-Eastern region of India. It is a multivoltine insect with 5-6 generations a year.
- The Ahom kings patronized muga silk industry in Assam. *Persea bombycina* and *Soalu* (*Litsea polyantha*) are the important food plants of the muga silkworm.
- Fully grown tree of 12 to 20 years can support the rearing of 5 to 10 layings and yield 500 cocoons in one season and one tree can be utilized for two rearing in a year.

Their life cycle is similar to that of mulberry silk insect and completes in about 50 days in summer and 120 days in winter. Like the mulberry silk insect, the muga silk insect also has four stages in the life cycle - egg, larva, pupa and adult.

LIFE CYCLE OF MUGA SILKWORM

EGG:

Each female moth lays around 150 eggs. The egg is oval in shape, dorsoventrally flattened, bilaterally symmetrical and creamy to brownish grey in color and measures 2.3-2.8 mm × 2.4 mm. They hatch in about 7 days in summer and 16 days in winter.

LARVA:

The newly hatched larva is black in color with distinct yellow lines at the inter-segmental region. The body tubercles are yellow and are provided with setae. The head is small but distinct and black in colour. They measure 0.6-1.3 cm in length.

They actively feed on the leaves of their food plants and pass through four moults like the mulberry silkworm and reach a size of 4.0 - 5.5 cm in length and weighs approximately 4.10 - 5.15 gm.

PUPA:

The dorsal surface of the Male Moth Mating pairs Eggs Female with Eggs on khorixha Larva-I Larva-III Larva-IV Larva-V Larva-II Female Pupa Cocoon.

Types of Silkworms body are light green while the ventral surface is deeper in colour. The fully grown larva spins a cocoon around it and then transforms into a pupa.

Spinning of cocoon takes 3 to 7 days in different seasons. The pupal stage lasts for 14 days in summer and 40 days in winter. The population in the hills undergoes diapauses at the pupal stage to overcome the winter.

ADULT:

Moths emerge from the cocoon, mate and lay eggs and continue their generation. Like the mulberry silk moths, they do not have mouth parts and die in about a week.

3. ERI SILKWORM

This silkworm derive its name 'Eri' from 'Eranda' in Sanskrit or 'Endi' in Assamese languages which means castor its main food plant. The domesticated species, *Samia ricini* is multivoltine, while, the wild species are usually trivoltine.

Castor

LIFE CYCLE ERI SILKWORM:

This silkworm derives its name 'Eri' from 'Eranda' in Sanskrit or 'Endi' in Assamese languages which means castor its main food plant. The domesticated species, *Samia ricini* is multivoltine, while, the wild species are usually trivoltine.

The life cycle of Eri silkworm has four stages - egg, larva, pupa (Cocoon) and adult (moth). A complete life cycle lasts about 44 days in summer and 85 days in winter .

EGG:

The eggs are ovoid in shape, candid white in colour, measures 1.5 x 1.0 mm and weighs 6 mg. By the time they hatch, they become ash to blackish colour. They hatch in 9-10 days in summer and 14 - 15 days in winter.

LARVA:

The new born larva is greenish yellow in colour measuring about 5 x 1 mm. The fully grown larva measures about 7.0 x 1.5 cm. and weighs 8 g, translucent and covered with a white powdery substance. Different strains have different colours and body marking pattern. The larval span varies from 20 days in summer to 50 days in winter. Eri moth 2nd, Instar larvae,, Eri silkworm eggs, , Eri cocoons, 5th, Instar larvae 12.

PUPA:

Fully mature larva spins a cocoon around it and then transforms into a pupa. Spinning takes around two days and pupation takes another two days. The cocoons are long, soft and without a peduncle. One end is narrow and open. Pupation usually takes around 10-20 days depending on the temperature.

ADULT:

The adult moth emerges from the cocoon in about 10 days from the day of spinning. After mating, each moth lays around 400 eggs which depend on the feed and the season. The total life cycle lasts about 45 days in summer and 90 days in winter.

The life cycle of Eri silkworm has four stages - egg, larva, pupa (Cocoon) and adult (moth). A complete life cycle lasts about 44 days in summer and 85 days in winter.

4. TASAR SILK WORM:

A. mylitta is the tropical tasar silk insect reared largely in central India and *A. proylei* and *A. roylei* are temperate tasar silk insects reared in the Himalayan belt. They belong to the family Saturniidae of the order Lepidoptera.

EGG:

The egg is oval in shape, dorso-ventrally flattened and bilaterally symmetrical along the antero-posterior axis. Eggs of tropical tasar measure about 3 mm in length and 2.5 mm in diameter and weighs approximately 10 mg, while, oak tasar, eggs measure about 2.5 x 2 mm and weighs about 7 mg.

Freshly laid eggs are dark brown in colour. After washing, it becomes white, light yellow or creamy. The eggs hatch in around 12 days from the day of hatching.

LARVA:

The newly hatched larva is dull brownish yellow with black head, measuring about 7mm long and 1 mm diameter and weighs about 8 mg. The larva at maturity measures about 13 cm long and 2.1 cm diameter and weighs about 50 gm. Various types of tubercles are present on the body. There are several hairs and setae, which are white, minute and irregularly distributed over the body. Shining spots are also present at the base of dorsal tubercles.

The tasar silkworms moult four times and pass through five instars. The larval period varies between 30 and 70 days during different seasons in different races. The fully grown larva spins a cocoon taking support of one or two leaves and Egg, Larva, Cocoon, Adult. Types of Silkworms forming a hammock and a peduncle to firmly cling to the plant. The process takes place in 4-6 days and the shell is very hard.

PUPA:

The pupa is the resting stage in the tasar insects. Pupal period varies greatly lasting from days to months depending upon the voltinism.

MOTH:

Females are bigger (4.5 cm) with a broad abdomen and narrow bipectinate antennae of 1.5 cm long. Males are smaller (4.0 cm) with a narrow abdomen and broad antennae. The females are gray or yellow, whereas, males are brown or yellow or gray. Mouthparts are reduced, as moths do not feed. In male, the wingspan is about 16 cm, while, in female, it is about 18 cm. Post-median line (PM) is red with a white line on the border. Ante-median line (AM) is usually black or dark brown in colour. The ocellus (OC: 70mm) with a transparent area is prominently positioned at the centre of the wing. Each female moth lays around 200 eggs which continue their generation.

VOLTINISM

Voltinism is the ability of silkworm to produce one to several generations in a year. Based on Voltinism the silkworms are broadly classified into three types.

(A) UNI-VOLTINE

(B) BI-VOLTINE

(C) MULTI OR POLY-VOLTINE

(A) UNI-VOLTINE

- Silkworms producing one generation in a year are called uni-voltine.
- Uni-voltine lays diapause eggs due to absence of juvenile hormone.
- They have a long life cycle, Larvae and Cocoons are large. They are very sensitive to environmental conditions.
- They are unsuitable for summer and autumn.
- The diapause eggs are reared by artificial breaking.
- The cocoon filament is of good quality.
- The shape of cocoon is round and oval which is white and pale yellow in colour.

(B) BI-VOLTINE

- They produce two generations in a year.
- The first generation adult lays non-diapausing egg.
- The second generation adult developing from non-diapausing egg laid during first generation lays diapause eggs due to absence of juvenile hormone, which is dormant till next spring.
- The larval duration is short. Larvae are robust and tolerate environmental conditions.
- The leaf cocoon ratio is less.
- The quality of the cocoons is inferior to that of Uni voltine races. Cocoon weight, shell weight, silk percentage and filament length are lesser than uni-voltines.
- Cocoons are dumbbell or oval in shape, white or pale yellow in colour.

Example: NB4D2, NB18, KA, NB7 etc.

(C) MULTI-VOLTINE

- They produce more than 5-6 generations in a year.
- The larval duration is short.
- The leaf cocoon ratio is high. Cocoons are small in size. The cocoons produce inferior quality silk than uni and bi-voltine races. The shell ratio is less.
- The filament length is short. The filament is fine and clean with little lousiness, but with more lustrous.
- The larvae are robust and can tolerate fluctuating environmental conditions and hence best suited for tropical climates.
- They lay only non diapausing eggs due to presence of juvenile hormone. Example: Pure Mysore, C.nichi, Hosa Mysore.



MULBERRY CULTIVATION



Mulberry cultivation and sericulture have been practiced for a very long time. But in the past mulberry was cultivated more for its multipurpose use than for rearing of silkworms alone. After the establishment of the Central Silk Board, emphasis has been laid on the raising of mulberry exclusively sericultural purposes.

Mulberry is a perennial plant. Hence, for maximum and sustained yield under given environmental conditions, a suitable variety of mulberry must be recommended. In this choice of variety, assistance is given by the technical and field assistants of the State Governments.

LEAF SELECTION

Mulberry for late age worms are also selected by largest glossy leaf method which is described in chapter 2.5. In glossy leaf method selection, the mature leaves remained on the mulberry twigs after chawki rearing are fed to IV and V instar worms plucking from top to bottom.

IMPORTANT MULBERRY VARIETIES OF TAMILNADU

Mysore Local and M-5 are the two most common varieties use in Tamil Nadu. Mysore Local can grow in rain-fed conditions also as it is drought-resistant. It can grow quickly and also withstand climate variations. M-5 is a superior hybrid variety evolved by CSR&TI. Mysore and gives a higher yield. Further, the leaves are of a better quality with more protein content and give a higher leaf-cocoon ratio. However, it needs supplemental irrigation and fertilizer application.

In addition to these, some new varieties have been introduced in Tamil Nadu and are cultivated at the farmer level. Some of these are S-30, S-36, S-41, S-54 and MR-2. Of these, S-30 and S-36, though better than m-5 are poor rooters. The drawback of S-41 is that it matures earlier than the other S-varieties. Therefore, S-54 is considered to be the best of the series. It is suitable for irrigated conditions.

ADVANTAGES:

1. It has got large leaves.
2. Its leaves have a smooth waxy surface.
3. It has shorter intermodal distance.
4. It has quick sprouting potentially.
5. Its leaf productivity under different eco-climate conditions is superior .
6. It has more crude protein, free amino acids, total minerals and total soluble sugars than K-2. 7.

At the farmer's levels, feeding S054 resulted in higher cocoon yield than M-5. Dry food ingested to produce 1 am of silk by NM18 is 11.5 gm for S-54, 13.5 for K-2 and 16.3 for Mysore Local. But its disadvantage is that it is not suitable for free plantation, as its branches droop down. Mr-2 is a mildew resistant variety evolved by CSRS, Connor.

It is popular among the Rasipuram taluk of Salem district. It has more crude protein, free amino acids, total minerals, total soluble sugars and lipids than M-5 and is suitable for free plantation also.

Some of the other newly-evolved varieties worth mentioning are V-1 or Victory evolved by CSR&TI, Mysore with a yield potential of 67 MT per hectare and a high photosynthetic efficiency and consequently more proteins, sugars, etc. It also has a high efficiency of water use.

It is suitable for both Bivoltine, and multivoltine into Bivoltine hybrid rearing. SV1 is a new somoclonal variety evolved from S-1 through tissue culture and is expected to yield 38 MT/ha/yr, and is under field trial.

PREPARATION OF MULBERRY CULTIVATION

1. TYPE OF PLANTATION:

The plantation could be of the tree, middling or bush type. S-1301, S-36, S-523, TR-4, Ber-C-776, S-41, S-30, TR-8, M.laevigata, Local, K-2 and Ber-S-1 are the varieties recommended for tree plantation, Kanva 2, Local and others suited to the local conditions for bush, while no specific variety has been recommended for the middling type so far..

2. TYPE OF SOIL:

S-1, S-1635 and S-799 are the varieties recommended for the alluvial soil of the Gangetic plains, C-776 and S-1 for saline or alkaline soil and C-776 and C-763 for late rite soil.

3. AVAILABILITY OF WATER:

S-1, C-763 and S-795 are suggested to be ideal for the heavy rainfall regions of West Bengal and Assam. The improved varieties viz., D-54, S-41, S-36 and S-30 are considered suitable for irrigated conditions. S-61, S-34, S-14 and S-13 considered suitable for rain-fed conditions have not yet made their appearance in the field S-13, S-14, S-763 and C-776 are the drought-tolerant varieties recommended for Orissa.

4. PREPARATION OF LAND

Mulberry being a perennial, land is prepared initially. The field is leveled first. It is ploughed deeply initially using heavy mould board plough, up to a depth of 12" to 15" in order to loosen the soil. Weeds and gravel are removed.

A basal dose of organic manure like compost or cattle yard manure is applied at the rate of 10 tonners per hectare for rain-fed and 20 tons per hectare for irrigated land. The manure is incorporated by repeated ploughings. This enables easy establishment of the crop initially.

5. PLANTING DIRECTION AND SEASON:

Planting direction is of no consequence in tropical countries where plenty of sunlight is available. In temperate regions, it is recommended to plant the material in a direction facing sunlight and parallel to the direction of wind.

Planting must be done in a season favorable for the growth of seedlings. It must be avoided in winter and summer. Early spring and late autumn are considered ideal for planting in temperate countries. In India, the recommended planting season is the beginning of the monsoon period.

Planting in the middle of the rainy season will result in the rotting of the cuttings. July-August, when south-west monsoon begins are ideal months for planting in Karnataka and Tamil Nadu. In West Bengal, November is the recommended period as north-east monsoon is over by then.

6.PLANTING DISTANCE

The inter-plant and inter-row distance varies depending upon many factors, Chief among which are:

1. Whether mulberry is grown as a pure, mixed, companion or hedge crop:
2. Soil fertility:
3. Intensity of cultivation:
4. Agro-climatic conditions:
5. Method of training, harvesting and
6. Availability of soil water.

Maintaining a high plant density will give increased yield per unit area when climatic conditions and agricultural inputs are not acting as limiting factors.

In general, wider spacing is adopted only under rain-fed conditions. With the exception of Jammu & Kashmir, where mulberry is grown as a tree and branch harvesting is done, in other states close-spacing is practiced.

The inter-plant and inter-row distance recommended for tropical countries is 3' * 3' for rain-fed and 2'*2' for irrigated crop and for temperate regions like Kashmir (tree crop), it is 4'*4'. The spacing followed in the southern states, where intense mulberry cultivation is practiced is 1'*1.5', 2'*2' and 3'*3'.



PLANTING SYSTEMS

This system followed for rain-fed crop adopts a wider spacing. Instead of ploughing the entire field, pits of standard size (40*40*40 cm) are dug with an inter-plant and inter-row distances of 90*90 cm if it is the bush type of cultivation, 180*90 cm if it is to be high bush and 270*270 cm if it is to be a tree plantation.

TYPES OF PLANTING SYSTEMS:

- 1. ROW SYSTEM**
- 2. KOLAR SYSTEM**
- 3. STRIP SYSTEM**
- 4. ANGULAR SYSTEM**

1. ROW SYSTEM:

This system is followed for irrigated mulberry crop throughout South India. This land is prepared by ploughing well. And ridges and furrows are made by ploughing. The distance between the ridges is generally 45-60 cm. A rope with knots at equal distances of about 45-60 cm is tied from one end of a ridge to the opposite end, and two cuttings are planted at the point indicated by the knot on either side of the ridge.

2. KOLAR SYSTEM

This is similar to the row system except that the distance between the plants is very much less. The spacing between the rows is between 30 and 45 cm and the distance between the plants in the row is 10-15 cm. This is followed in the Kolar district of Karnataka and so is called the Kolar system.

3. STRIP SYSTEM:

This very close system of mulberry cultivation is practised in West Bengal, where mulberry is grown in strips. Each strip has either two rows (Dothaki) or three rows (Thethaki). Each strip is separated from the adjacent strip by a wide distance so that harvesting and inter-cultivation and other operations can be done using small tractors or other machines. Within the strip, plants are planted at a distance of 15 cm between the rows and between two plants within the row.

4. ANGULAR SYSTEM

This is a new system of cultivation evolved by RSRS, Coonoor for the slopes of the Nilgiris. The distance between the plants is similar to the pit system. i.e., 90 cm, but the plants in adjacent rows are planted in such a way that they form a triangular arrangement. This system is claimed to allow more plants per unit area, better soil and moisture conservation, efficient cultural operations, increased leaf production, better supervision and efficient and economic water management.

IRRIGATION

The quality and quantity of leaves harvested from mulberry plantation for silkworm feeding depends upon the availability of soil water. Only in areas where there is uniform rainfall at the rate of 100-150 mm per month throughout the year, the leaf production of mulberry not limited by the availability of water.

METHODS OF IRRIGATION:

1. Furrow Method:
2. Flat Bed Method:
3. Basin Method
4. Overhead Sprinkler Method:
5. Micro-irrigation Methods:



TYPES OF IRRIGATION SYSTEMS

There are many different types of irrigation systems, depending on how the water is distributed throughout the field. Some common types of irrigation systems include:

1 .SURFACE IRRIGATION:

Water is distributed over and across land by gravity, no mechanical pump involved.

2. LOCALIZED IRRIGATION:

Water is distributed under low pressure, through a piped network and applied to each plant.

3. DRIP IRRIGATION:

A type of localized irrigation in which drops of water are delivered at or near the root of plants. In this type of irrigation, evaporation and runoff are minimized.

4. SPRINKLER IRRIGATION:

Water is distributed by overhead high-pressure sprinklers or guns from a central location in the field or from sprinklers on moving platforms.

5. CENTER PIVOT IRRIGATION:

Water is distributed by a system of sprinklers that move on wheeled towers in a circular pattern. This system is common in flat areas of the United States.

6. LATERAL MOVE IRRIGATION:

Water is distributed through a series of pipes, each with a wheel and a set of sprinklers, which are rotated either by hand or with a purpose-built mechanism. The sprinklers move a certain distance across the field and then need to have the water hose reconnected for the next distance. This system tends to be less expensive but requires more labor than others.

7.SUB-IRRIGATION:

Water is distributed across land by raising the water table, through a system of pumping stations, canals, gates, and ditches. This type of irrigation is most effective in areas with high water tables.

8.MANUAL IRRIGATION:

Water is distributed across land through manual labor and watering cans. This system is very labor intensive.

MANURING

Mulberry requires a total of 15 elements, of which C, H, O, N, P and K constitute the major nutrients as they are required in large amounts. Ca, Mg and S are next in importance and constitute the secondary nutrients. Fe, Mn, Cu, Mb, B and Cl are the minor nutrients. The requirement for cobalt and zinc in trace amounts has now been established.

The two routes of entry of these elements are:

1. The roots and
2. The leaves.

Generally C, O is obtained through the leaves as CO₂ from air and the rest through roots from soil water.



ORGANIC MANURES USED IN MULBERRY

1. COMPOST:

It is made by the decomposition of grass or straw or other domestic and agricultural wastes in pits under anaerobic conditions. Lime-milk is added to quicken decompositions. Compost is like well-decomposed cattle manure in general appearance, more powdery and light in colour. The nutrient content in compost and FYM have the same properties and value (Table 5.1).

2. FARM YARD MANURE (FYM):

It consists of dung, urine and straw that has been used as bedding for cattle. Fresh FYM is harmful to mulberry roots. In tropical regions, due to high temperature, decomposition is very quick, and partially decomposed FYM can be used.

3. SERICULTURE WASTES:

In Japan and China all the wastes generated during the cultivation, pruning, harvesting and rearing of silkworms are recycled as mulches and fertilizers to the mulberry garden. A small amount of superphosphate of lime is added for decomposition. The wastes thus used as fertilizers after decomposition include, silkworm litter, dead worms, exuviate, pruned leaves, pupal wastes, and dead and unused months.

4. POULTRY MANURE:

The wastes generated in poultry farming consisting of litter, unfed feed, and dropped feathers are rich in phosphates and come under the concentrated category. They are good fertilizers but have to be used only after decomposition.

5. OILCAKES:

Oilcakes of soybean, neem and groundnut are used in India though they generally come under the concentrated category of organic manures. Of these, neem cake, especially, is recommended for mulberry as it is said to have nematocidal properties.

6. GREEN MANURES:

Leguminous plants which have nitrogen-fixing root nodule bacteria are grown as intercrops in mulberry fields and, after their yield is harvested, are mulched and used as green manures. Plants that have been grown thus are sun hemp or Crotalaria, Indigofera, Cajanus, Phaseolus, Acacia and others. Now, neem and Karaj leaves at the rate of one ton per acre per crop are used as mulches.

7. BIOGAS SLURRY:

As biogas plants liberate only the bound chemical energy and leave the nitrogenous material of the organic matter used undiminished, the slurry of biogas plants are ideal manures for mulberry as well as other crops.

8. PRESS MUD:

It is an important by-product of the sugar industry. It contains considerable quantity of calcium, organic matter and phosphorus along with nitrogen, potassium and certain trace elements. Press mud is profitable phosphate manure for mulberry. Application of 7.5 tons of press mud per

hectare is equivalent to 150 kg of P_2O_5 , single superphosphate per hectare. Press mud is available at sugar factories at a nominal price, sometimes even free of cost.

ADVANTAGES OF ORGANIC MANURES:

- Increases the water-holding capacity of the soil.
- Improve the physical condition of the soil. Organic manure results in better aggregation of soil particles so that it is rendered easy to handle and also it facilitates the passage of air and water.
- Besides the direct effect, organic manure also has a residual effect in the next crop.
- Increases the chemical and biological activities of the soil.
- When used in combination with chemical fertilizers, enhances their efficiency.
- Minimizes the volatilization and leaching losses of nutrients.
- Application as mulch on soil helps in the conservation of soil moisture.
- Imparts a dark colour to the soil and helps the soil to absorb heat from the sunlight and regulate the soil temperature.
- Increases the availability of native plant nutrients in the soil.
- Makes complexes with the nutrients and releases them slowly.
- Imparts a dark colour to the soil and helps the soil to absorb heat from the sunlight and regulate the soil temperature.



CHEMICAL FERTILIZERS

1. NITROGEN:

This element is the most important to be considered in fertilizer application because it is present in soil in small amounts, while its content in the leaves is about 1.5-3.5%. Further, soil nitrogen is soluble and is easily leached away.

Four forms of nitrogenous fertilizers are available.

1. Water soluble proteins and other organics.
2. Water insoluble organic non-proteins substances.
3. The Nitrates and
4. The Ammonia

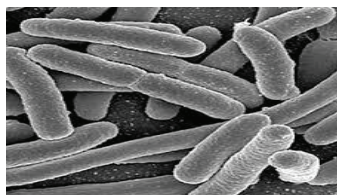
2. PHOSPHORUS:

Phosphorus deficiency can be recognized from stunted growth and very small dark-green leaves. Older leaves appear reddish-purple. Phosphate fertilizers occur either in the water-soluble form or in the citrate soluble form. Important water soluble phosphorus fertilizers are simple or triple super phosphates.

3. POTASSIUM:

Soil, potassium is essential for sustained yield of quality mulberry leaves as it is required for the formation of the woody parts of the plant. Potassium is generally applied as potassium sulphate, potassium met sulphate or muriate of potash. Of these, the last is highly hygroscopic and must be stored carefully.

BIO-FERTILIZERS



Any biological material or process used for adding essential nutrient content to the soil is called a bio-fertilizer. However, normally the term bio-fertilizer is restricted to the inoculation of specific micro-organisms to the soil for the purpose of enriching soil nutrients by their activity. The microbes are mass cultured and marketed as Bio-fertilizers. Widely used Bio-fertilizers for mulberry include the following.

1. AZOTOBACTER:

This is a free-living heterotrophic nitrogen-fixing bacterium. It is adapted for neutral to alkaline soil. It increases the yield of non-leguminous crops by 5-15%. Apart from nitrogen-fixing, it also produces growth-promoting substances like IAA, Gibberellins (GA), B Vitamins and antifungal substances. The contribution of nitrogen to the soil by this bacterium is about 25 kg/ha/yr. Another important characteristic of this bacterium is that it excretes ammonia in the modification of nutrient uptake by the plant.

2. AZOSPIRILLUM:

This is a slightly curved, Gram-negative, rod-shaped bacterium exhibiting a spiral movement. It is an aerobic form, storing polyhydroxy butyrate as its reserve material. It fixes nitrogen only under micro aerophilic conditions. This nitrogen-fixing bacteria is found in loose association with the roots, utilizing the exudates released by the plants as its nutrients. It returns, it fixes atmospheric nitrogen for the host. This relationship is called associative symbiosis.

3. AZOLLA:

This is an aquatic fern which is used as a biofertilizer because of its symbiotic nitrogen-fixing blue green alga. The advantage of using this aquatic plant is that it decomposes quickly and makes the nitrogen available very efficiently and also checks the growth of other weeds.

4. VAM:

These are some species of fungi growing as mycorrhiza on the roots of the plants. These are called vesicular borescent mycorrhiza because of the nature of their growth. The popular VAM species used as biofertilizer is *Glomus fasciculatum*. This is used for phosphate utilization rather than for improving nitrogen availability.

ADVANTAGES OF BIOFERTILIZER:

1. Low cost of production and better input-output ratio.
2. Total absence of harmful effects.
3. Environmental safety
4. Repeated inoculations not required once the optimum populations are built up in the soil
5. Leaching loss is minimum owing to slow release of nutrients and inherent recycling mechanisms.
6. Feasibility of multiplication and population build-up is there.
7. Biofertilizer often produce more than one effect. For example, nitrogen fixers also make phosphates available and, in addition release growth promoting substances.

LIMITATIONS OF BIOFERTILIZER

1. Poor cross adaptability of the strains and technology among different crops.
2. Slow impact on crop productivity owing to slow release of nutrients.
3. Non-feasibility of meeting higher levels of nutrient requirements.
4. Performance unpredictable and ranges from 0 to 65% increase in yield.
5. Inoculums' is susceptible to unknown biotic and a biotic factors of the environment and may even fail to establish.

PRECAUTIONS:

1. Chemical fertilizers should not be mixed with Bio fertilizer unless specified.
2. Bio fertilizer should be used before their expiry date.
3. Bio fertilizers should be collected from authentic sources, preferably from Government agency or a university to avoid spurious material.

PRUNING AND TRAINING



Periodically some branches of the mulberry plant are methodically cut off, and this operation is called pruning, and is as important as fertilizing or tilling for ensuring a good yield of quality leaves.

TYPES OF PRUNING:

There are four types of pruning. They are:

- a) Bottom pruning or low-cut pruning, in which branches are cut to the base;
- b) Middle pruning, in which the branches are cut around mid-height at about one metre;
- c) Top pruning or high-cut pruning, in which the branches are cut at the top or to the soft portion and
- d) Kolar system, in which the branches are cut to the ground level, wherein the pruning and harvesting are done together.

MERITS OF LOW-CUT PRUNING:

1. Harvesting and preparation of cuttings are easy, because the stumps are straight.
2. Harvesting rate increases because sprouting occurs easily and it can also be done early.
3. It has high utility value in temperature regions because the hardening of the leaves is delayed and growth is rich even during late autumn silkworm rearing.
4. Damage due to pests and diseases is less, and when they attack, it is easy to take control measures.

DEMERITS OF LOW-CUT PRUNING:

1. Since the age of the plant is shortened, dwarf disease occurs easily.
2. Plants are susceptible to damage by frost, snow and Dogare disease.
3. During heavy rainy season, mulberry leaves get soiled easily.
4. It is recommended for warm regions and regions where the soil is shallow and there is rich ground water. It is unsuitable for heavy rainfall regions and flood-prone areas.

MERITS OF HIGH-CUT PRUNING

1. The age of the plant is longer and the quality of the leaves is relatively better.
2. Dwarf disease is less. Damage due to frost is less. Less susceptibility to Dogare blight.
3. The water content of the leaves is less and they mature easily.
4. Catch cropping is easy.
5. Less damage due to floods.

DEMERITS OF HIGH-CUT PRUNING

1. Difficult to harvest leaves
2. Difficult to manage stump adjustment
3. Plant more susceptible to pests and diseases
4. It takes too long a period for the plants to get ready for harvest.
5. Harvesting rate is also lower.
6. Plants get easily damaged by wind and it is difficult to supplement for dry stump..It is recommended for cold regions, rainy and flood-prone regions.
7. The number of pruning in a year varies from place to place. For rain-fed crop, pruning is given only once a year in July. August just prior to the monsoon. For irrigated crop, particularly the Kolar system, pruning is done five times.

CARE AT PRUNING:

1. The cuts are injurious and care is to be taken to minimize the damage.
2. The cut should be clean without any cracks.
3. The bark round the cut should be intact without any peels.
4. Sharp, good and ideal implements should be used.
5. Lime washing of cut ends is to be done to prevent entry of pathogens.
6. Application of fungicides, wax coating, and oil application to the cut ends should be done to prevent entry of fungal pathogens.
7. When there are two annual basal pruning, the second can be made to the mid-level of the branches or slightly lower and this reduces one injury to the plant.

METHODS OF HARVESTING

The nutritive value of the leaf changes according to the photosynthetic and respiratory activities of the leaf. Leaves harvested in the afternoon contain less water and more of carbohydrates due to the active photosynthesis and transpiration taking place in day time. Such leaves wither more quickly and hence it is recommended that the leaves are harvested in the early hours of the morning.

The method of harvesting leaves varies in different areas according to the rearing method used and is modified to suit the intensity of rearing and availability of labour at the given time. Whatever the method, the following leaves are to be avoided for silkworm feeding: buds, too tender leaves, dusty leaves, diseased and pest-infested leaves, insecticide sprayed leaves, declining, over-mature and coarse leaves.

1. LEAF PICKING

Leaves are picked individually from the main stem with petioles. At the same time, the terminal buds are nipped off so that lateral shoots develop rapidly, and sufficient leaves are available for the next crop. Leaf picking starts ten weeks after bottom pruning and subsequent pickings can be done at intervals of about seven to eight weeks.

2. BRANCH CUTTING

This method called 'Batchi' system in Kashmir is also practised in West Bengal and parts of Karnataka in India, and also in Japan and Russia. In this method, the entire branch is harvested and used to feed worms after the III moult directly. Normally this is used with tree and middling plantations and shoot and floor rearing.



3. WHOLE – SHOOT HARVEST

This method of harvest consists of cutting the branches to ground level by bottom pruning and feeding the entire shoot to larvae after IV moult. Shoots are harvested at intervals of 10-12 weeks and thus 5-6 harvests are made in a year. As shoots are cut to the ground levels, the shoots sprouting reach uniform maturity by the next harvest.

This method is popular in Kolar district in Karnataka and Malda district in West Bengal in India. In Japan and Russia, whole shoot harvest is done with mechanical harvesters.



PRESERVATION OF LEAVES:

- Fresh mulberry leaves contain a lot of starch which is not palatable to silkworms. Further, generally only one or two harvests are done per day while worms are fed four to five times per day.
- A certain time must necessarily lapse during the transport of leaves from the fields to the rearing room.
- During this period from harvesting to feeding, mulberry leaves are preserved and stored. Properly preserved leaves cut down labour on feeding by reducing the number of feeds.
- During the period of storage, the chemical constitution of the leaves is altered because of metabolic activities.
- The most visible change is withering from water loss. Water is lost by transpiration and hydrolysis reactions while some water is produced during respiration. However, there is a net loss of water which is proportional to the time of storage.
- In the initial period of storage, starch is broken down to sugars, the increase of which leads to a higher nutritive value and palatability of the leaves to the worms. But with the lapse of time, the sugars are oxidised during respiration and are depleted.
- Proteins are also hydrolysed to amino acids. However, there is no net decrease of total nitrogen.
- The crude ash and fibre content are not affected during storage period.
- The major problem during storage is water loss and deterioration of the nutritive value by the breakdown of carbohydrates and proteins.
- The rate at which this occurs can be reduced by storing the leaves below 20°C and over 90% R.H.
- Mulberry leaves for infant worms may be stored in baskets but those for grown-up worms require storage in larger wooden or bamboo leaf chambers covered on all sides with a wet gunny cloth.

- In view of the importance of storage for better cocoon yield, it is recommended that leaves be kept in small leaf preservation bags of 2 * 3 sizes, having a capacity for holding 3-4 kg leaves.
- The bags, made of gunny cloth or polythene sheet or both, are double-layered.
- Such small bags are more convenient to handle and easier for storing large amounts of leaves for a longer duration than the traditional method of storing in heaps and covering them with wet gunny cloth.



PESTS OF MULBERRY



Mulberry is a perennial tree which sprouts throughout the year in the tropical countries. It offers food and shelter to a number of insect and non-insect organisms. Mulberry is periodically pruned and the leaves are harvested for feeding silkworms, pest attacks on mulberry are sporadic and seasonal, and most of the pests are polyphagous. They may be classified as:

- 1) Leaf-eating pests,
- 2) Sap feeding pests and
- 3) Stem borers.

A. LEAF-EATING PESTS (lepidopteron pests)

1. SPILOSOMA (DIACRISIA OBLIQUE)

- This is also called Black-headed or Bihar hairy caterpillar.
- It is a polyphagous pest affecting castor, groundnut and jute. Mulberry is also one of its host plants.
- Young caterpillars feed gregariously on the under surface of the leaves and skeletonise them. From the youth instars onwards, the larvae disperse and defoliate the plants leaving only the main veins. They cause serious loss to the Rearers.

CONTROL MEASURES:

- Physical control methods include the collection and destruction of eggs and gregarious larvae and practising deep ploughing and flood irrigation to kill the pupa in the soil.
- Spraying of 0.2% DDVP is a common chemical method and the leaves are to be harvested after a safe period of 3 days.
- Biologically, the pest can be controlled by releasing the larval parasite *Apanteles oblique*.

2. EPISOMUS FUGULUS

- This is a leaf-eating coleopteran pest which belongs to the family Curculionidae.
- This pest damages both old and new mulberry plantations. In severe infestation, the entire plantation is often given up due to the death of the plants by pest damage.
- This has been experienced in the Malda district of West Bengal. The highest incidence of the pest occurs in spring.
- The weevil affects the plant in the grub and the adult stages. The grub is more detrimental though it remains on the under surface of the soil at a depth of 28-45 cm because they feed on the cortical tissue of the roots.
- This affects the ascent of sap in the roots and the plant wilts and dies.
- The grubs may also move upwards and feed on the shoots of the young plant. The adult feeds on the leaves and lays eggs on the under surface of the leaf.

Control measures:

- These are mainly directed towards killing the underground larval stages. Deep digging, food irrigation and soil solution are some of the suggested mechanical methods of control.
- Application of Aldrin dust (10 kg/acre). Thimet-10G (3kg/ha/yr) and Neem oilcake (1500 also controls the larva.

4. MIMA STRA CYANURA

- This is a coleopteran pest belonging to the family Chrysomelidae.
- This pest is generally distributed in the subtropical climate and is, therefore, prevalent in the Jammu & Kashmir state of India.
- It is polyphagous pest feeding on a number of forest trees including almond, fig, wild pear, apricot and plus.
- In addition, mulberry is also a host plant for this insect.

5. OTHER BEETLE PESTS

- Other beetles which have been recorded from mulberry include – Apion ampulum, Altica himensis, Luperomorpha nigripennis, Monolepta signatu, Chirida bipunctata, Cryptocephalus schestedti, Aspidomorpha miliaris and Raphidopalpa abdominalis.
- The adult stages of these beetles make holes in the mulberry leaves by their feeding activity.

Control measures:

- Hand-picking and destruction of egg and larvae and spraying of 0.2% DDVP (Dimethyl Dichloro Vinge Phosphare) or Dimethoete effectively controls the pest..
- Leaves of the insecticide sprayed plant should be used for silkworm rearing after three days.

6. ORTHOPTERAN PESTS:

- The grasshoppers are polyphagous pests feeding on anything green and are general defoliators.
- The grasshoppers that have been recorded as pests of mulberry in the different parts of the country include the long-horned grasshoppers, *Letana inflata* and *Pheneroptera gracilis* and the short-horned grasshoppers, *Ailopus sinulatrix*, *Cyrtacanthacris ranaceae*, *Attractomorpha crenulata* and *Colemania sphearioides*.

7. DIPTERAN PEST:

A Dipterans shoot gall midge called *Diplosis mori* has been found eating the growing parts of mulberry without forming a gall. It is found in Japan.

8. SNAILS:

Some terrestrial snails have been recorded as defoliators of mulberry leaves in certain parts of India in certain seasons. These include *Chochiopa* sp., *Cryptozona semirugata* and *Machrochlamys pedana*.

9. SAP FEEDERS:

- Apart from hemipteran bugs, thrips and mites also feed on the juice of mulberry plant and affect its growth.
- These generally have piercing and sucking mouth parts adapted for feeding on plant juice.

C. HEMIPTERAN PESTS:

1. EMPOSCA FLAVESCENS (LEAFHOPPER OR JASSID)

- These are small greenish leafhoppers which suck from the veins on the underside of leaf. "Hopper burn" is reported to be due to a virus introduced by the hoppers.
- Appearance of triangular brown spots at the tip and end of each vein is a symptom of leafhopper bite.
- The affected leaf rolls and becomes scorched. Both nymphs and adults feed on the plant.

Control measures:

Include sprinkler irrigation, setting up light trap. And spraying of 0.05% DDVP.

2. PSEUDODENDROTHRIPS MORI

The leaves affected by this thysanopterans show white streaks which become yellowish-brown on maturity. The colour of the nymph is pale green, of the adult male is brownish-yellow and of the females is dark brown.

Control measures:

It includes adopting sprinkler irrigation and spraying 0.02% DDVP at weekly interval.

3. TETRANYCHUS EQUITORIUS AND T. TELARIUS (Mites) (Order: Acarina)

The mites spin white silky nets on the underside of leaves. They suck the sap of the leaves. Lightly infested leaves have pale spots on the leaf. Heavily infested ones have pale-rusty spots and dry up.

Control measures:

Sprinkler irrigation, release of coccinellids and application of Parathion aerosol or sulphur dust are some of the control measures used.

4. BORER PESTS

These bore into the stem and their larvae tunnel through and feed on the stem tissues. The stem dries and breaks resulting in poor leaf yield. They have powerful mandibles for cutting the wood

5. STHENIAS GRIATOR (STEM GIRDLER)

These beetles girdle young, green shoots at night. Grubs can girdle through dry wood.

6. XYLOTRECHUS CHINENSIS:

This beetle is a Japanese pest and has not been recorded from India.



SILK REELING

(Raw Silk Manufacture)

The process of drawing silk fiber from the cocoon is called 'Reeling'. The cocoons are cooked in hot water and the silk fiber is unwound from the cocoons. The silk consists of two proteins, the inner core of fibroin and an outer cover of gum Sericin.

During reeling, the cocoons are processed in hot water at 95-97°C for 10-15 minutes. This process is called cooking. This cooking will enable the Sericin portion to get softened and make unwinding easy without breaks. The cocoons after cooking are reeled in hot water in different types of machines. In India, 61% of the silk amounting to 1,320 tones is reeled on the country-type charka (spinning-wheel) numbering 6,656.

The silk produced with the country charka is of very poor quality, as the thread is not uniform; as it carries many slubs etc. the improved cottage-type basins have been introduced recently into India. Provision for button-holes and a proper crosier system to maintain the thickness of the fiber, and to control the defects of neatness in the cottage basin have facilitated the production of better-quality silk. As much as 4,000 cottage basins in the country produce 806 tons of silk.

Large-scale basins organized scientifically are arranged in filatures for the extraction of superior-quality fiber. The silk produced by the filatures is superior because of the low level of defects of cleanliness and uniformity in the thickness of the fiber. Only 8% of the total production of silk in India is contributed by filatures.

The silk produced from multivoltine races of silk worms is poor in quality and is known to have greater defects, such as lousiness, and defects in neatness and cleanliness and is of very poor quality in light of the international D grade. The silk produced by the Bivoltine races of silkworms possesses superior neatness and cleanliness, is without lousiness and has high tensile strength and stands to the international A grade.

The new technology of handling silkworms in the country has shown that Bivoltine silkworms, producing the international quality of silk, can be produced throughout the year in the Karnataka state in India. It has been shown that, on an average, 30 to 35 kg of cocoons, yielding 3 to 4 kg of high-grade silk can be produced by rearing 40,000 eggs of Bivoltine, as compared with 25 kg of cocoons and 1.4 kg of low-quantity fiber from multivoltine.

FACTORS AFFECTING THE QUALITY OF SILK:

1. WATER

2. COCOON DRYING

1. WATER:

- The water used in silk reeling plays an important role. It is estimated that to produce one kg. Of silk from charka, the volume of water required is about 100 liters.
- If the quality of water is poor, the impurities suspended in the water will adhere to the silk thereby the colour, and Lustre of silk would be influenced.
- The water used for cooking cocoons should be colourless, odour less and transparent. The pH of water should be between 6.8 to 8.4.

2. COCOON DRYING:

- The cocoons contain live pupae which will emerge out within 10-12 days of cocooning hence, cocoons are stifled.
- Once the moth emerges out, the shell becomes of no use to reel since the adult emerges out by piercing the cocoon shell.
- The cocoons contain live pupae, which will emerge out within 10-12 days of cocooning hence, cocoons are stifled.
- Once the moth emerges out, the shell becomes of no use to reel since the adult emerges out by piercing the cocoon shell.
- The primary object of drying cocoons is to kill the pupae and to reduce the water content of fresh cocoons.

METHODS OF STIFLING:

1. SUN DRYING

1. The freshly harvested cocoons are directly exposed to sunlight.
2. In this method the harvested cocoons are spread in one layer and directly exposed to bright sun-light for several days from morning to evening until the pupae inside is killed and cocoon is found properly dried.
3. This can be identified by shaking the cocoon which will make rattling sound.

ADVANTAGES:

1. This is cheapest method of stifling
2. Easily available method of cocoons drying to Rearers.
3. These cocoons can be reeled immediately.

DISADVANTAGES:

Continuous exposure to hot sun-light, damages the silk quality due to the action of ultra violet rays.

1. Sometimes continuous sun-light may not be possible, particularly during rainy season. Partial drying of cocoons will spoil the quality of silk.
2. Wastage of silk during reeling will be more.

2. STEAM STIFLING METHODS

Steam stifling is the very common and easy method.

- The fresh cocoons are directly exposed to hot wet steam.
- Basket steaming
- Barel steaming
- Chamber steaming

ADVANTAGES

- i. conveniently used to stifle large quantities of cocoons
- ii. uniform stifling of cocoons

DISADVANTAGES

- i. steam stifled cocoons cannot be reeled immediately, otherwise the sericin will wet and fibroin comes in lumps while reeling
- ii. in steam stifling pupa is killed inside and does not dry it.

3. HOT AIR CONDITIONING:

In the modified hot air conditioning chamber there are five compartments. Each chamber is different ranges of temperature.

- The first chamber is filled with cocoons are supplied of with hot air at 50°C for about 3hrs.
- The second chamber by sliding mechanism by pulling out where 70°C for about 3hrs.
- The third chamber cocoons are dropped into $75-80^{\circ}\text{C}$.
- The fourth chamber is cocoons dried by hot air at $80-90^{\circ}\text{C}$ temperature
- The fifth chambers are gradually exposed to hot air in the 12 hours of time and continuous process.

ADVANTAGES

- i. Any amount of cocoons can be easily stifled.
- ii. Time is saving, labor saving.

DISADVANTAGES

- i. Cocoons of uniform size are quite suitable for this process, but Indian multi-voltine cocoons with different sizes and different qualities need different degrees of temperature for stifling.
- ii. Costly and skilled labors are required.

4. STORAGE OF COCOONS:

While storing the cocoons certain precautions should be taken into consideration otherwise improper storage of cocoons leads to loss due to occurrence of damage to the cocoons.

PREVENTIVE MEASURES SHOULD BE TAKEN:

1. Avoid storage of stained cocoons
2. Storage room should have good ventilation.
3. Cocoon storage place should be clean and dry. No waste cocoons should be stored in the premises.
4. Always ensure proper drainage of cocoons before storage.
5. Relative humidity in the storage room should not exceed 70%.
6. Before storage of cocoons, the storage place should be thoroughly disinfected.
7. In case of long storage of cocoons, frequent turnings should be given.

5. SORTING

Sorting of cocoons is already done at once, once again it should be checked prior to reeling.

6. DE FLOSSING

Floss is the outer most layer of cocoon which is only a protective covering and is not reelable. Prior to reeling floss has to be removed by any of the methods to avoid obstruction during reeling.

7. COCOON RIDDING

After de flossing the cocoons, in the process of ridding, cocoons are separated according to their sizes by simple sieves into big, small and medium size cocoons and collected in separate containers. This will facilitate the reeling operations easy.

TYPES OF SILK REELING

Reeling is a process of unwinding of silk filament from the cocoon. Reeling process is an important activity. Depending upon the required thickness (denier) of silk thread filaments from number of cocoons are combined together and reeled.

An efficient Reeler will maintain the fixed number of cocoons per end to produce uniform denier silk. Reeling is carried out by distinctive methods.



CHARKA

It is about 50% of total raw silk production is contributed by charka. Reeling machine is traditionally home built by using wooden material with the assistance of black smith and carpenter hence,

- Investment on reeling machinery is very-very less.
- Even the inferior quality cocoons can be also reeled by charka.
- No much technology is involved in this system of reeling.
- Only one reeler can reel.
- Brushing and cooking are done in the same basin.
- The cost of production is very low compared to other reeling machines.
- Involvement of labour is very less only two persons are sufficient to maintain.
- Reeler can establish a charka unit at the back yord of his own residence.
- Production per day would be around 1 kg from each charka.
- Marketing of charka reeled silk is very easy and it will be always at his door steps.

In charka, there are three distinct parts.

1. Mud-Platform
2. Distributor
3. Reel.

ADVANTAGE

1. The reel should be always smooth.
2. As the cooking and brushing, reeling are done in the same water. The water should be changed frequently.
3. As the cooking and brushing, reeling are done in the same water.
4. After cooking the cocoons thoroughly, when reeling ends are found, the temperature of water in the reeling basin should be lowered to 40 °C by adding cold water.

DISADVANTAGES

1. The silk produced by charka reeling is inferior quality.
2. As the reeling is done in water at very high temperature which is harmful to the reeler.
3. Frequently, water is to be changed and heated again, this will increase the fuel cost.

- **COTTAGE BASIN**

In cottage basin, the cocoons are cooked separately and re-reeling is done separately to increase the quality of silk. In cottage basin cooking is done separately and the reeling basins are fixed to a reel bench. There will be 6-10 ends in each basin. Arrangement to supply hot water to each basin and to drain out the dirt water accumulated due to floss and sericin is made.

ADVANTAGE

1. Quality of silk produced is superior to charka silk.
2. Water in the reeling basin will be fresh. This will help to maintain the quality of silk.
3. Production per day can be increased, as it is power operated.

POINTS TO BE REMEMBERED BY REELER:

The waste material stuck to the jute about should be removed from time to time.

1. It is advisable to take a mixture of fresh and partially reeled cocoons while reeling.
2. Replace the water in the reeling basin from time to time.
3. Apply the lubricant to crosser pulleys for smooth running.

- **FILATURE**

Filature or multi-end reeling machine works on the principle of slow motion reeling and thread production on small reels at a large number of ends per basin.

ADVANTAGES:

1. Superior quality silk can be produced
2. Wastage is reduced
3. High productivity
4. Less labour
5. Arrangements are made to control the speed of the reels and tension of the threads.
6. Raw silk per day will be around 800 gms of superior quality.

AUTOMATIC REELING MACHINE:

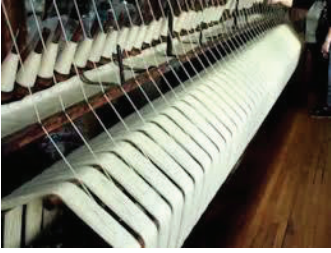
- Bi-voltine cocoons are best suited to automatic reeling machines.
- This machinery requires superior quality cocoons and uniform size of cocoons with less floss.
- Generally the multi-voltine cocoons are not fit for automatic reeling.
- This is particularly due to inferior quality.
- The silk produced from automatic reeling machine will be superior.

RE-REELING

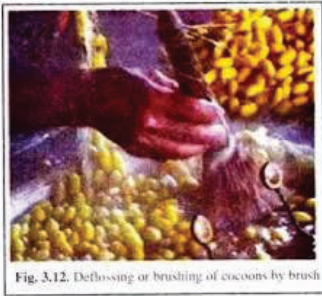
- Sometimes silk after being reeled on device like charkha or cottage basin, may suffer from some defects like short length, broken threads, entanglements, hard gum spots, etc. These problems may create wastages of thread during subsequent throwing and twisting.
- To overcome these defects silk is first reeled on small reels and later is transferred to standard sized hanks. This process is called re-reeling.
- When silk is directly reeled on standard reels as in charkha, it suffers from certain defects like short lengths of fine sizes, broken threads, entanglements, hard gum spots, etc. These make wastages during the later operations of silk throwing and twisting.
- To avoid these defects, most reeling units practice indirect reeling where the actual reeling is done on small reels and the silk is later transferred to standard sized hanks and this process is called re-reeling.
- In this process the raw silk on the reel is soaked in water to find the end, and then small reels are arranged on the reel stand.
- The re-reeling machine is provided with independent traverse rod with guide hooks for six ends to be re-reeled onto six standard sized grant reels.
- The defects in the silk are removed during re-reeling. Skeins of acceptable length, width and weight are made from the standard reels.
- The strain given to the silk thread during reeling is released during re-reeling. Each reel is provided with a hand brake.
- In some re-reeling machines, stem pipes are also fitted below the reels to dry the raw silk during re-reeling.
- The weight of the skein is kept at 70 gm up to 33 denier and 140 gms for above 34 denier raw silk.



RE-REELING PICTURES



LACING, SKEINING



- The two ends of the re-reeled silk hank are tied with a coloured silk thread; the colour varying with the denier reeled.
- To keep the diamond pattern of the hank from being disturbed, the hank is divided into five parts and is laced with a cotton thread.
- The lacing also helps in the prevention of entangling of the silk in later operations.
- As a preliminary step to packing, the reeled hank is converted to a skein. Before skeining, the hank is examined in a silk examination room under diffuse light to remove defects remaining by oversight.
- The cleaned and laced hanks are put on a skeining machine to be skeined. During the process, the hank is twisted and folded upon itself to form a number of tight spirals. This prevents ruffling and entanglement of the filaments during packing.
- The skeins are made into books. In a book there are eight skeins in the horizontal row and five in the vertical row.
- Each book is neatly tied with separate cotton bands at three different places and wrapped in tissue paper.
- The books are then packed into bales of twenty kgs. The bales are stored in humidity-free, air-tight and fire-proof rooms till they are sent to markets.
- Re-reeled silk hunk is then tied with a coloured silk thread by the two ends. The hunk is then divided into five parts and is laced with a cotton thread.
- This lacing prevents entangling of the silk in later operation.

BOOK MAKING:

The cleaned and laced hunks are then put on a skeining machine to be skeined. During this process, the hunk is twisted and folded upon itself to form a number of tight spirals or skeins the skeining prevents ruffling and further entanglement of filaments during packing.

The skeins are then made into books. In a book, eight skeins are placed in horizontal row and five in vertical row. Each book is then tied with separate cotton bands at three different places and wrapped with tissue paper.

Books are kept in bales of twenty kg weight. Thus, silk threads are kept in store house as bales. Store house should be free from humidity and air.



Fig. 6: Raw silk books from CSR bivoltine hybrid cocoons

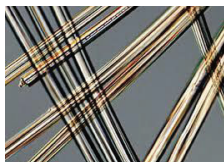
EXAMINATION OF SILK (COCOON QUALITY AND CLASSIFICATION)

COCOON QUALITY TESTING:

A series of natural circumstances will produce variations in cocoon quality. Some of the most noteworthy include:

- Differences in cocoon quality in the same batch;
- Differences in cocoons produced in the same location by different farmers who have reared the same species;
- Seasonal influences. In Japan for example, cocoons produced in the spring and late autumn are higher in quality than those in early autumn and summer;
- Environmental conditions affect cocoon reel ability such as temperature and humidity;
- Processing technique in reeling will impact reeling efficiency as well as raw silk quality;
- Bivoltine cocoons are superior quality compared to multivoltine silkworm species traditional farmed in tropical zones.

Recent silkworm cultivation now develops cross-breeds of multivoltine with bivoltine silkworms as a strategy to improve overall cocoon quality.



FACTORS INFLUENCING COCOON QUALITY:

This section presents the measures to be taken during silkworm rearing and mounting to obtain a better quality of cocoons with higher silk content, longer filament, better reel ability and lower percentage of defective cocoons.

1. TEMPERATURE AND HUMIDITY DURING MOUNTING

Maintain temperatures at or near 25°C and relative humidity around 65 percent for silkworms to spin good quality cocoons with high reliability.

2. MOUNTING DEVICE

Although different mount practices are employed among producer countries, rotary mounting frames provide good ventilation. The result is improved reliability of cocoons.

3. HARVESTING AND HANDLING OF FRESH COCOONS

Cocoons should be harvested only following complete pupation. In practice, the appropriate harvesting day would be the fifth day in tropical countries, and the seventh or eighth day in temperate countries, from the mounting date. If premature harvesting takes place, the silkworm will still be in its larval stage, weigh more, have fragile skin, and could likely be crushed, which would cause stains to the cocoon during handling and transportation.

4. TRANSPORT OF FRESH COCOON

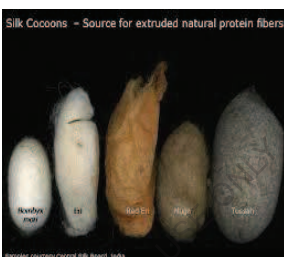
After proper harvesting and removal of diseased or damaged cocoons, the fresh cocoons are taken to the market. For short distances, the farmer carries the cocoons in bamboo baskets or jut bags on his head or by bicycle. If the distance is longer, cocoons are transported in a van or a bus. Caution should be exercised when loading fresh cocoons on to the van to ensure that containers are loosely packed in tiers to avoid damage. Vibration and shock during long trips can spoil fresh cocoons. Cocoon quality is affected by steam produced while being transferred in a bag or basket.

If there are defective cocoons fresh cocoon quality will be harmed. Describes the impact of transportation on fresh cocoons. While it is advisable to avoid carrying cocoons over long distances, there are steps, which preserve silk reliability.

First use of P.V.C. containers with 15 kgs capacity is recommended. Shock absorbers, such as sponge can prevent damage over long distances.

To minimize the risk of heat deterioration, shipping should take place only during the night or early morning. Ideally, the fresh cocoons should arrive at the stifling unit within two to three days after harvest.

CLASSIFICATION OF COCOONS:



When cocoons are sold at the market, price is assessed on the basis of cocoon quality. This is judged by grading shell percent, filament length, reliability and the percentage of defective cocoons.

If the percentage of defective cocoons is high, the price will be affected. The next section outlines the characteristics of defective cocoons.

DEFECTIVE COCOONS:

1. DOUBLE COCOONS

A double cocoon is spun by two worms, producing a filament, which does not unwind smoothly and tangles easily. As these cannot be reeled along with normal cocoons, double cocoons are used for manufacture of a coarse, non-uniform, stubby yarn called "doupion". Double cocoons may be caused by crowded mounting conditions, high temperatures, high humidity and mutation of silk species.

2. INSIDE STAINED COCOONS (DEAD COCOONS)

Dead cocoons are also known as melted cocoons. In this case, the pupa is dead and sticks to the inside shell of the cocoon causing a stain. Melted cocoons are called mutes because they do not make a sound when shaken. These cocoons are difficult to process and will result in silk, which is dull in colour.

3. OUTSIDE STAINED COCOONS

These are recognized by a rusty colour spot on the cocoon shell caused by absorption of intestinal fluid/urine of the mature worm formed during mounting. Reel ability is very poor in this case.

4. PRINTED COCOONS

This defect may happen due to improper mounting frames; these are also called scaffold pressed cocoons.

5. MALFORMED COCOONS

These are abnormally shaped cocoons, which may arise from species variation. This defect may be due to racial characteristics and breeding with mulberry leaves stained with agrochemicals.

6. FLIMSY COCOONS

Here, the shell is loosely spun in layers and has a low silk content. These cocoons are easily overcooked and produce waste.

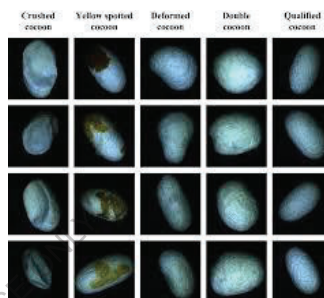
7. THIN-END COCOONS

One or both ends of the cocoon are very thin and risk bursting when processed. The cause of this defect may be attributed to species characteristics or improper temperature and humidity during rearing and mounting.

8. PIERCED COCOONS

This happens when a moth has emerged, been eaten by beetles or in the case of the emergence of a parasite. Pierced cocoons are unfit for reeling and can be used only for hand spinning or as raw material of machine spun silk yarn.

COCOON TESTING AND GRADING



- In sericulture advanced countries, cocoons are subjected to systematic testing and grading before sale.
- Prices are based on the quality of the cocoons. But in developing countries, there is no system for cocoon testing.
- Cocoons are sold on visual inspection and personal experience is relied upon in marketing of cocoons.
- No laws exist or compulsory testing and trading of cocoons.
- The result is that cocoons are simply auctioned or in certain instances, even sold at a price fixed by the Regional Departments of Sericulture.
- There is no direct correlation between price and quality of cocoons.
- Major sericulture countries through the International Sericulture Commission (ISC) have studied an International Cocoon Classification System. However, diversion of silkworm species, techniques of breeding, silk reeling and other factors lead to the production of non-uniform cocoons.
- These parameters have challenged the effort to unify equipment and testing methods

- Cocoon testing and grading may be accomplished with a compact automatic reeling machine as well as a multi-ends reeling machine, which is typical equipment in major sericulture countries.
- Testing methods used by major sericulture countries are displayed.

The quantity of fresh cocoons, which are taken out of a lot for testing purposes, depends on the actual weight of the lot on offer.

The cocoons on offer are divided into three batches.

- 1.Batch weighing up to 1 000 kg**
- 2.Batch weighing up to 2 000 kg**
- 3.Batch weighing up to 4 000 kg**

The sample size of fresh cocoons taken out of each batch for testing is as follows:

- **1st batch - 2.0 kg**
- **2nd batch – 4.5 kg**
- **3rd batch – 6.0 kg**

In the case of dry cocoons, the quantities taken out for testing from each batch are as follows:

- **1st batch up to 400 kg of dry weight – 0.8 kg taken out for testing.**
- **2nd batch up to 800 kg of dry weight – 1.8 kg taken out for testing.**
- **3rd batch up to 1 600 and over - 2.4 kg taken out for testing.**

1. DRYING OF THE COCOON TEST SAMPLE.

The cocoons received must be dried as soon as possible and to an acceptable degree. To do this, the moisture content of the fresh pupa and the cocoon shell percentage must be measured. The percentage of drying is calculated according to the formula.

$$\begin{aligned} \text{Drying percent of cocoon} &= (0.0115 \times \text{mc of fresh pupa} - 0.2104) \\ &\quad \times (\text{percent of cocoon shell } 1.15) \\ &\quad \times (\text{mc of fresh cocoon} + 115) \\ &\quad (\text{mc: moisture content of the cocoons}) \end{aligned}$$

Drying should be accomplished in one continuous process where the temperature is gradually decreased from 98°C to 60°C until the required ratio is obtained.

2. TEST TO CALCULATE PERCENTAGE OF ELIMINATED COCOONS

This evaluation usually happens on a table under natural light. If natural light is insufficient based on time of day or weather conditions, elimination takes place under artificial light of 500 Lux.

To calculate the percentage of eliminated cocoons according to terms and conditions defined by cocoon classification, the following types of cocoons must be removed: double cocoons, thin-end cocoons, scaffold marked cocoons, malformed cocoons, flimsy cocoons.

3. BATCHING OF COCOON SAMPLED FOR REELING

After elimination of the bad cocoons, the remaining cocoons are batched for the reeling test as follows:

Batch	Sample cocoon (fresh weight)	Cocoon for reeling (dried weight)	Preliminary cooking
	Kg	grams	grams
1 st	2.0	300 x 2	80
2 nd	4.5	300 x 2	80 x 2
3 rd	6.0	300 x 3	80 x 2

4. COOKING OF THE COCOON SAMPLE

A small sample of cocoons must be cooked to determine the correct cooking conditions for the specific batch. Once these parameters are established, actual cooking of the entire sample batch can be completed.

5. REELING OF COCOON SAMPLE:

Reeling should be carried out under the following conditions:

Item	Multi-end reeling	Automatic reeling
Temperature of groping end part	85°C	80°C
Reeling velocity	90 m/min	160 m/min
No. of reeling silk ends (per basin)	10 p cs.	3 p cs.
Length of croissure	10 cm	8 cm
No. of cocoons or objective size per reeling thread	8 pcs.	21 denier

6. RE-REELING

This should be carried out on large reels with a circumference of 1.5 m and the standard re-reeling speed of 160 r.p.m.

CALCULATION OF RESULTS

The results of at least two or three reeling tests must be taken in order to calculate the classification. The resultant raw silk weight divided by the sample cocoon weight will indicate the raw silk percentage. The cocoon classification items are worked out by the following method:

1. PERCENTAGE OF COCOON SHELL

Points to be observed for testing percentage of cocoon shell are as follows:

- The percentage of the cocoon shell can be calculated from the weight of the whole cocoon and cocoon shell obtained from 200 cocoons from which 100 cocoons are taken separately.
- If possible, equal amounts of female and male cocoons should be selected for the sample.
- No defective cocoons should be included in the sample.
- As atmospheric conditions make the moisture content of the cocoons too variable, the sample should be selected from a point 10 cm beneath the surface of the batch of cocoons.

The average value of the percentage of the cocoon shell obtained from repeating the test twice must be graded within a 0.3 percent deviation. But if the difference between both samples is + 0.3 percent another test should be taken.

2. ESTIMATED COCOON PERCENTAGE

The result should be expressed to one decimal place where the weight of sample cocoons is the sum of eliminated cocoons plus the weight of the good cocoons.

3. LENGTH OF COCOON FILAMENT

Where, Total reeled cocoon numbers=

Sample cocoon number – Converted carry over cocoon number

Length of raw silk is checked by the gauge. Sum of reeling cocoon number per thread is the total number of reeling cocoon verified 20 times during the reeling (this means once per unit work process). The length is based on the average of every reeling block.

It is expressed by total number to the one decimal place. Converted unreel able cocoon number and carry over cocoons are calculated by converting them into length and expressed by full cocoon number to the one decimal place.



SILK WORM REARING

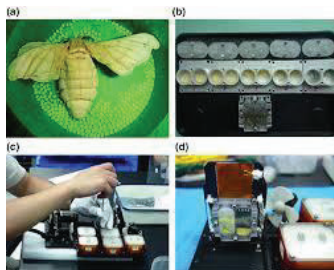
Silkworm eggs are of two types i.e. hibernating and non-hibernating eggs. Further processing of the eggs depends upon whether they are of the diapausing or the non- diapauses type. Uni voltine races lay only diapausing eggs. Multi voltine races lay only non-diapausing eggs while the behaviour of the eggs of the bivoltine is intermediate. Except multi voltine, uni and bi voltine race eggs are hibernating (diapausing) eggs which require special treatment to make them hatch.

These eggs are stored till the next season or awakened from diapause artificially. The eggs stored are taken out and subjected to incubation to achieve uniform hatching on a desired day. This can be achieved by exposing the eggs to certain range of environmental conditions. The incubation of eggs is one of the essential parameter in silkworm rearing.

1. INCUBATION:

- The quality of eggs plays a vital role in silk worm rearing.
- The eggs are to be incubated properly which ensure maximum hatching percentage.
- Incubation is a phase in protecting the activated silkworm eggs before rearing.
- Incubation means to provide suitable environmental conditions i.e. temperature 25oC +1oC and humidity 70 to 80% for uniform development of embryo to achieve good hatching percentage.
- The environment influences the stored eggs in the development of the embryo, uniform hatching of the worms, and health of the larvae, mortality and yield of cocoons.
- Thus the silk worm eggs are kept under proper incubation process to get high and uniform rate of hatching on the desired date, good larval health and high cocoon quality.
- The cold preservation treatment, acid treatment (artificial incubation) and followed by ideal incubation process are keen factors for achieving good quality cocoon crop.
- If any unexpected reason, for which the hatching has to be delayed after incubation has started, can be done to a limited extent by cold storing the eggs at 50C on the second or third day of incubation for about a week.
- Hatching can be delayed at blue egg stage by cold storing for about a week at 50C. Brushing can also be delayed by cold storing the newly hatched larvae for about three days at 7.50-100C.

2. HANDLING OF EGGS:



- The incubated eggs are handled properly to get good hatching percentage; otherwise harmful effects may result in poor or un uniform hatching.
- Silkworm eggs are produced in central places, viz., by the Government grainages or private licensed seed producers.
- Rearers have to purchase the eggs from egg producing centres only.
- The eggs being exposed to live environmental conditions, which will result unhealthy development of embryos. Hence, there is a need to transport eggs properly to ensure healthy and uniform development of the eggs.
- Live silk worm eggs are transported from egg producing centers (grainages) to the Rearer's rearing places or chawki rearing centers (young silkworm rearing centres) with proper care, because during transport also it is important to maintain optimum temperature and humidity.
- Eggs exposed to high temperature (above 280 C) and humidity (over 90%) during egg-laying and incubation; storage of hibernating eggs at 250 C for too long period or under dry and high temperature conditions; stimulation of eggs in acid treatment; contamination of eggs by pesticides, nicotine, mosquito repellent, glue or gum causes rotating of eggs before head pigmentation.
- Rotten eggs are dead eggs which perish after the colouring of the serosa and before head pigmentation with sunken shell and deep cavity. Too high temperature (over 280 C) during incubation; to low relative humidity (less than 50%); contact with pesticides or other harmful chemicals causes dead eggs after head pigmentation. In this embryo is normally formed in the eggs shell but death occurs before hatching stage.
- This keeping un view of the above dangers the eggs are to be handled carefully till hatching so as to ensure uniform and more hatching.

3. HATCHING:



- Silk worm eggs are available loosely or on egg cards.
- The newly developed larva breaks out the eggs shell and come out, and is called hatching.
- The hatched larvae are brushed and reared. The newly hatched larvae are black, hairy and look like small ants and are called “ants” or “kego”.
- It is always better to brush the larvae in the morning. The eggs exposed in early hours or hot hours results in irregular hatching.
- Generally brushing should not be delayed. If necessary, can be postponed to next day when hatching is irregular.
- Brushing should be completed in the morning/cool hours of the day. Freshly emerged larvae can also be preserved at 7-100C for one day.
- The ratio between hatched eggs and total no. of eggs in a laying is called “hatching percentage”.
- The hatched eggs, unfertilised or dead egg number is counted individually for calculating the percentage.
- This can be done using a different coloured ink pens usually red and later it is calculated using the following formulae and method.

Total No. of eggs hatched X 100

Total No. of eggs

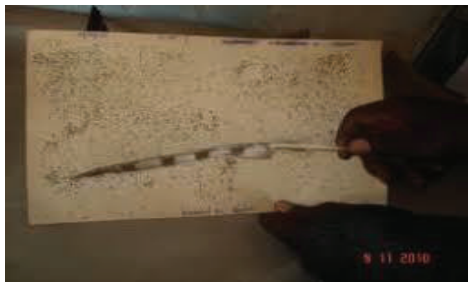
Total No. of dead eggs X 100

Total No. of eggs

Total no. of eggs

= Good eggs+ dead eggs.

4. BRUSHING:



- When the eggs hatch, the emerged larvae are to be collected for rearing.
- This process of separating from egg sheet or egg shell is called “brushing”.
- Which involve sprinkling of thinly chopped mulberry leaves on just hatched larvae to attract them to feed.
- The larvae which crawl on to leaves will be separated and transferred to rearing tray from their shells by using optimum time for brushing is around 10 am when the humidity is 83 – 90% and temperature is 27*c.
- Normally hatching starts at 5 – 6 am when the eggs are exposed to early sun rays. Uniform hatching can be expected by 7-8 am.
- After 2 hours the newly hatched worms develop appetite and begin to crawl.
- Thus the suitable time for brushing would be 10 am but once again it depends upon the weather conditions.
- Before brushing, rearing facilities should be prepared according to the number of silkworms to be reared.
- Tender mulberry leaves should be chopped to 3-5 mm size. When the worms crawl over in to the tray worm bodies should be disinfected for disease prevention by using a fine-mesh sieve to dust a fine powder of formalin (3%).

METHODS OF BRUSHING

(Brushing of eggs is of two types depending on type of eggs).

I. Brushing of loose eggs.

II. Brushing of egg cards.

I. BRUSHING LOOSE EGGS

- The eggs are spread evenly in one layer in the egg box and kept in black box at blue egg stage. On the next day when all eggs reach blue egg stage they are removed from black box and covered with a thin perforated cloth or a fine mesh or finely perforated paper.
- This covering is placed in such a way that it just touches the upper surface of the eggs. Then just before brushing, chopped mulberry leaves are sprinkled on the top of the net or cloth or paper.
- This mulberry leaf attracts the hatched worms to crawl on to the upper surface. When maximum number of worms hatches out and crawls on to the paper they are transferred to rearing tray.

II. BRUSHING EGG CARDS

There are four methods,

(a) TAPPING METHOD

- The hatched larvae which crawl towards the edges or corners of the egg sheet are collected in to rearing tray using a feather.
- The egg sheet is held upside down just above the rearing seat and tapped.
- The larvae fall on to the rearing seat.
- Then the dropped larvae are brushed together with feather and fed with mulberry leaves which are already chopped.

(b) FEATHER METHOD

- The hatched larvae which crawl towards the edge or corners of the sheet are collected gently with white feather.
- Hold the egg sheet slantwise to the rearing seat, transfer larvae with the help of a feather.
- Later worms are fed with finely chopped mulberry leaf.

(c) BRUSHING WITH MULBERRY LEAVES

- Mulberry leaves chopped to 0.5 sq.cm size are sprinkled on the egg sheets when larvae hatch out.
- The mulberry leaf attracts the larvae.
- After 10 minutes the egg sheet is turned upside down and larvae on mulberry leaf are transferred to rearing tray using feather.

(d) HUSK - FEEDING METHOD

- Finely powdered paddy husk is sprinkled, evenly over the eggs sheet when the larvae hatched.
- Then finely chopped mulberry leaves are sprinkled over the worms which crawl on to leaf are transferred to rearing tray by using feather.

(e) NET AND FEEDING METHOD OF BRUSHING

- In this method spread a net on the hatched larvae then sprinkled chopped leaves over the net.
- The leaves attract the larvae. After 30 minutes leaves, worms along with net transferred to another rearing tray.

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SILK WORM REARING TECHNIQUES

- The life cycle of silk worm consists of egg, larva, pupa (cocoon) and adult stages. Among these four stages, larval stage is the only feeding and active stage.
- The duration of larval period from hatching to spinning is about 26 days. During this long duration the larvae grow in size and enter cocoon (pupal) stage.
- To accommodate the larval body growth the larvae undergo four moults and the complete larval duration can be clearly differentiated into five instars or stages.
- The first three instars (till the third moult) are known as young age or chawki and the last two instars are called as late age worms.

QUALITY OF MULBERRY LEAF:

- The mulberry leaf is the exclusive food of the silkworm *Bombyx mori*.
- The growth of the silk worm very much depends on the quality of leaves fed to them.
- The leaf quality is influenced by various factors such as soil, pruning, fertilizer, rainfall, irrigation etc., with these conditions mulberry grows luxuriously with rich contents of proteins and carbohydrates.
- Further the leaves are also succulent due to high nutrient content. These types of leaves are edible for silk worms for better growth and to produce good cocoons, leaves of Mulberry grown on loamy soil contain more water, protein and less carbohydrate and fibre, further the leaves mature slowly.

LEAF SELECTION:

- Mulberry leaf for young silkworms has a great effect on the growth and health of silkworm. Therefore selection of leaves should be done carefully.
- The leaves for young silkworms must be soft, tender, rich in water content, protein, carbohydrates etc.,
- There is high correlation between moisture content in the top tender leaves and chawki worm growth and moulting.
- Therefore for plucking the correct leaves for young worms, the largest leaf method is adopted.

FEEDING:

- Silkworms are fed to satisfy their appetite. Thereby uniform and healthy growth of silkworms can be achieved. For this, quality leaves are to be preserved and rearing beds are kept clean.
- Feeding with too much leaves is not economical.

MAIN OBJECTIVES OF FEEDING:

1. To satisfy the appetite of larvae
2. To promote eating and digestion of leaves by larvae
3. To keep the quality of leaves during eating.
4. To keep rearing beds clean
5. To avoid wastage of leaves and labour

Generally early age silkworms eat leaves from the surface, while late age worms from the edge.

The feeding activity of each instar silkworm can be conveniently divided into seven stages.

1. First feeding stage
2. Sparse eating stage
3. Moderate eating stage
4. Active eating stage
5. Pre moulting stage
6. Last feeding stage
7. Moulting stage.

GROWTH OF SILK WORMS:

Silk worms show high rate of growth. The growth by weight, between hatching and final spinning of cocoons stage is 10,000 times which is achieved in a matter of 24 to 25 days.

For achieving full growth of silkworm new techniques of rearing are followed by which worm grows to a weight of 4-5 gr.

These worms are healthy and produce cocoons of 1.75-2.0 gr. and above in weight. The growth of the worms mainly depends upon the amount of mulberry ingested and digested.

BED CLEANING:

Silk worms are fed with large quantity of mulberry leaves than their eating capacity. Thus unconsumed leaves which are unfit for food remain in the tray at the end of each feed. Besides these excreta of worms forms a thick bed. Out of the total weight of leaf has taken as food, three fifth is excreted and only two-fifth is being assimilated by the silkworm.

The piling of litter makes the beds moist. This releases process of fermentation liberating injurious gases and also favours multiplication of pathogens.

All these above factors are harmful to the worms. Therefore removal of old (unused) mulberry leaves, faecal matter of silkworms, exuviate, dead or unhealthy worms etc., from the rearing bed is called bed cleaning.

METHOD OF CLEANING:

For cleaning of beds husk, nets, cut straw are used.

There are three methods of cleaning.

A. CLEANING WITH HUSK

B. CLEANING WITH NET.

C. CLEANING WITH HUSK AND NET.

(a) CLEANING WITH HUSK

- For this method charred husk or paddy husk is sprinkled evenly over the bed of silkworms.
- This sprinkling of husk is carried just prior to first feeding early in the morning.
- The worms crawl through the husk layer to reach the leaves.
- During the second feeding the bed is ready for cleaning.
- All the worms are collected together by a brush and transferred into another fresh tray.
- The natural paddy husk is too big and too thick for two ages.
- Thereby the worms cannot come up.
- For these ages, husk should be broken into small pieces before it is used. Care should be taken to avoid dust of husk as it spoils the leaves fed to the worms.
- Formalized charred husk helps to avoid attack of Muscardine disease.

(b) CLEANING WITH NET

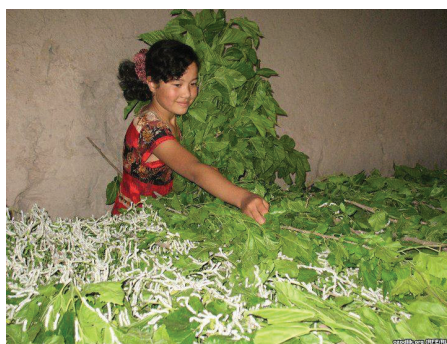
- In this process a net with mesh suited to the size of the worms are used.
- During the process of cleaning the net is spread over the bed just prior to the first feeding early in the morning.
- Then it is cleaned after second feeding.
- It is very simple method and requires little labour. However it is not convenient for the purpose of spacing.

The Mesh sizes of different cleaning Nets :

- First and second instar.....2mm²
- Third instar10mm²

(c) COMBINED HUSK AND NET METHOD

- In the process of cleaning both husk and net are used. First a thin layer of paddy husk is sprinkled over the bed and a suitable net is spread.
- Then after two feedings the worms are transferred along with the net into another tray.
- This process is more expensive and not suitable for spacing.



SPACING:

- This is an important aspect which needs maximum care. Overcrowded bed does not permit free and complete growth of the worms.
- It is very important for the vigorous and full growth of worms. As the worms grow in size and weight, the bed density increases leading to crowding.
- Therefore the population density in the rearing bed should be regulated to ideal condition.
- In rearing most of the failures are because of improper spacing in the bed. As the age of the worms increases the length and breadth increases.
- Overcrowding of worms means insufficient space for the movement and free feeding of the worms.
- Crowding condition favours to increase gases, heat and fermentation of faecal matter. Fermentation process particularly happens.

METHOD OF SPACING:

- The spacing can be done separately or in combination with bed cleaning. Among these two the latter method is convenient and satisfactory.
- It helps in less handling less disturbance of worms.



MOULTING:

- The silkworm larval life has five instars and four moults. The larva casts of its skin to accommodate the body growth this is called moulting.
- The silkworm larvae attain their maximum body growth of particular instar and as a result body becomes stout, and shiny and amber coloured.
- These two characters are seen in larva at the approach of moulting. In relation to the size of the body, the head of the worm appears small and dark.
- This is time for bed cleaning and wide spacing. After then worms are about to settle for moults are given one or two feeds which helps to reduce the humidity and favors uniform moulting.



SPINNING AND MOUNTING:



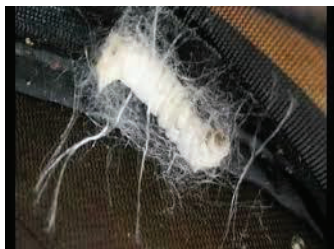
- The object of rearing silk worms is to get cocoons of good quality and maximum yield. Silk worm stops feeding at the end of fifth instar and starts building the cocoons.
- The larvae become transparent, shiny and stops feeding before spinning. Silk worm spins cocoons prior to pupation so as to protect itself from external disturbances and natural enemies.
- Since it is the most critical period of metamorphosis, good mountages help the rearer to get good quality cocoons.
- The cocoons are to be harvested carefully and cleaned to eliminate bad cocoons otherwise it reduces the cost of cocoons.
- The quality of cocoons is decided basing on shell weight, cocoon weight, shell ratio, floss percentage, no.of cocoons per kg., number of bad cocoons, filament length, number of breaks, denier etc.

RIPENING OF WORMS:



- Fifth instar worms feeding may last for five to seven days in case of multi voltine and bi voltine worms in the tropical areas, and seven to nine days in case of bi voltine and uni voltine races in sub-tropical areas.
- These worms stop feeding and called as mature larvae and starts spinning the cocoons. As the stomach contents become empty, the mature larvae become specific in appearance.
- They are translucent and yellowish with fully formed silk glands and it is a clear indication that the worms are fully ripe and ready for mounting.
- Ripe worms should be picked in time so all the mature worms are enabled to spin cocoons successfully.
- Worms not picked in time or unduly delayed in picking can also be mistaken as diseased worms.
- Worms picked much before ripening May not also spin, resulting in unnecessary crop losses at the last stage of rearing.

PROCESS OF SPINNING:



When the ripen worms are mounted on the mountages they pass out last excreta in semi-solid condition. When the humidity is high, excess body moisture is also eliminated as urine. After defecation ripe worm starts spinning the cocoon by selecting a suitable place in the mountage.

Each silkworm develops a pair of silk glands which synthesize silk by utilizing the amino acids coming from mulberry leaf.

The ripen worm anchors itself first to the mountage by oozing a tiny droplet of silk fluid which immediately hardens and sticks to the mountage. Then by swinging the head continuously the silkworm first lays the foundation for the cocoons by weaving a primary web.

The silk worm larvae moves its head in the shape of 's' or to spin the cocoon. The former shape is found in the outer layers of cocoon shell while the later type is usual in middle and inner layer. In this way the larvae forms layers of silk filaments around it and finally wrapped in a compact shell.

The first formed filament i.e. primary web constitutes the floss of the cocoons and is not reelable. The floss in Uni and bivoltine races is about 2 percent of the weight of cocoon. While in multi voltine more than 10 percent floss is seen.

The process of spinning continues about 1 to 2 days in multi voltines and 2 to 3 days in uni/ bivoltine. After the compact shell of the cocoons is formed the shrinking larva wraps itself in palade or gossamer layer.

Finally the larva detaches itself from the cocoons shell to transform into pupa or chrysalis. This layer (gossamer/Palade layer) does not form part of main shell. It is also not reelable and contributes to waste silk content.

MOUNTING:

Depending on the material and structure of the cocooning frames the number of bad cocoons may increase or much labour may required in mounting.

- Mature worms can be easily induced to cocoon spinning
- Total mountage space can be efficiently used.
- Manufacturing can be easily and cheaply undertaken.
- Durable structures.
- Storage attached to the mountage can be easily removed.
- Material used is resistant to high moisture or wetness.

TYPES OF MOUNTAGES:

Mature silkworms are collected and mounted on the mountages. This laborious job requires lot of skilled labor. The ripe worms are identified and picked by skilled labour and mounted on the mountages. This kind of mounting reduces the density of the mounted worms and incidence of double cocoons. Diseased worms can also be eliminated. Worms can also be mounted using nets or green branches.

Branches of green leaves are placed over the rearing bed and when the worms crawl on to them, they are taken out and shaken over a mat; dislodged worms are put on mountages. Similarly a net is placed over the bed after feeding mature worms, which are no longer feeding crawl upon to the nets are collected for mounting as in branch method. In shoot rearing early maturing larvae (1020%) are picked by hand and later remaining worms are collected by shaking the branches and later mounted.

METHODS OF MOUNTING:

1. PICKING – UP FOR MOUNTING

When one third of body of the silk worm becomes transparent, worms are picked up and put on the mountages. This method helps the rearer to mount the worms at right time. But labour expenses are high. Generally silkworms mature between 10 am and 3 pm. The worms become over mature producing cocoons of poor quality if the mounting is delayed with inadequate labour.

2. BY SHAKING THE SHOOTS

It is better for shoot rearing. When worms (5-10%) mature, they are picked up by hands for mounting. The remaining worms are left till maturity.

When 40-50% of the worms have matured mulberry shoots full of silkworms are shaken on paper, vinyl sheets placed on a mat.

The remaining worms are mixed with faeces and remaining mulberry leaf. Then these are covered with 23 layers of straw nets. After a gap of 30-40 minutes silkworms crawl up on to the net. these worms (roughly hundreds) are mounted on selected mountages.

Rotary mountages are leaned against the wall by putting 130-140 worms on each frame. More number of worms is put on the top frame than the lower frame so as to fill all the frames evenly when suspended.

ADVANTAGES:

- Even size cocoons
- Less floss
- No chance for double cocoons
- Less labour expenses

3. NET METHOD

A net is placed after feeding the silkworms. Mature (ripe worms) crawl onto the net while others feed. The net is separated along with worms for mounting.

4. BRANCH METHOD

It is similar to net method where only mulberry shoots are used instead of nets. When worms crawl onto shoots are mounted.

6. SELF – MOUNTING

First mature worms are picked by hand then a self-mountage frame is hanged on the surface of silkworm beds. Ripe worms crawl onto the mountage frame, which is later hanged. It reduces labour cost.

COCOON HARVESTING:



- The silkworm larva metamorphosis into pupa after spinning the cocoons for about 48 hours from the time they are mounted.
- Generally pupation takes place on the 4th day of spinning.
- Thus the worms inside the cocoons will be still in the form of pre pupa, which has a delicate cuticular skin.
- Thus if the cocoons are handled before this stage, the skin may rupture and body fluid will ooze and stain the cocoons, making it unsuitable for reeling.
- Thus early harvesting of cocoons should be strictly avoided.
- In course of time the pupal skin hardens and turns to dark brown. The cocoons are then harvested on the 5th day in summer and 6th day in cooler season.
- In the case of seed cocoons, they may be harvested on the 6th day. Harvesting must not be delayed beyond the above said period because it affects the reeling activity.
- Before harvesting the cocoons, the mountages are held in a slanting position with the cocoon side downwards and given a gentle shaking to dislodge the faecal material.
- The flimsy cocoons are taken out with forceps or chopsticks. Care must be taken not to rupture the cocoons as their body fluid or dead silkworms may stain good cocoons.
- The cocoons harvested are kept in thin layers in a tray or on a mat. After harvesting, the cocoons are first cleaned to remove pellets or debris sticking to the cocoons.
- Then they are sorted to separate bad cocoons.

TRANSPORTION OF COCOON:

- The cocoons should be put into loosely woven cotton bags each weighing about 10kg. They are loosely packed and transported in cooler hours of the day (i.e. morning or evening). If transported in larger bags, cocoons will be steamed or crushed.
- Strong vibration during transportation is harmful because cocoons might be crushed.
- While transporting cocoons are protected from direct sunlight and rain.
- It is also important that cocoon cost depends on the quality they are well protected from direct sunlight and humidity.
- Cocoons are placed in heaps samples are given for quality testing, basing on which cost is decided.
- Cocoons are transported to well develop areas of reeling to get good returns.
- The cocoon quality is based on certain aspects like hatching percentage, ERR percentages of good and bad cocoons, shell ratio, denier, filament length.
- All these factors influence the cost of cocoons.



GRINAGE TECHNIQUES

Grainage is one of the important aspects of sericulture. Therefore the Grainage operations directly reflect on survival rate, life span, growth, and quality of cocoon, etc. Hence it is necessary to conduct these Grainage processes with utmost care and technique.

All these processes must be recorded for future use to improve the skills, to identify the faults, to minimize the expenditure, to increase the production of know the Pebrine disease and to confirm improved varieties etc.

The silkworm eggs required for commercial rearing should be of high quality and free from diseases. To produce good quality seed there must be a good seed organization. Considering the importance of quality seed, the Sericulture countries have established a network of institutions for egg production and also to impart training to the staff working in Grainages.

Silkworm seed is divided into two types

- A. Reproductive seed
- B. Industrial seed.

A. REPRODUCTIVE SEEDS

These are used for producing the seed cocoons (parents of seed cocoons) which are required in large numbers for producing commercial seed.

The purpose of these is for maintaining the racial purity which is difficult to rear, So special care must be taken by technical staff. The selling price of these is 3050 % more than reeling cocoons.

Reproductive seeds are often multiplied in number in a series of breeding centers called breeding stations (P4 or P3, P2, and P1). In three or four stages in order to ensure that the racial characters are not diluted during the multiplication stages.

B. INDUSTRIAL OR COMMERCIAL SEED

There are specific hybrids between two or more pure lines of silkworm races and are reared by the sericulturist for producing cocoons on a commercial scale for reeling purpose. These are hybrid seeds, produced in special organizations called Grainages.

There are three aspects in seed production.

They are

1. Breeder's stock.
2. Basic seed multiplication.
3. Industrial seed production.

The race breeding stations are maintained by the government. These centres multiply great grand parents or the grandparents (P4 or P3) of commercial Grainage seeds. These centres contain all pure races, and are aimed to maintain purity of the races. These centers supply the basic or initial material for multiplication, to the breeding centers.

Normally the seeds of multi-voltine races (pure or CB) are multiplied in three-tier system (P3) while exotic pure breed races of bi-voltine, rare multivoltine have four-tier system(P4).The above said stations(P3 or P4) supply parental seed cocoons for rearing at foundation stock seeds(P2 Station).A multiplication of pure breed races are done by seed cocoon rearers who in turn produce seed cocoons.

These seed cocoons are brought to the industrial seed producing Grainages (P1) to produce the hybrid seeds which are also called foundation hybrids.

Generally Grainages where the laying of the parental races are produced and managed by government agencies whereas the industrial seed grainages may be private or government owned. These Grainages are aimed to produce reproductive and industrial seed.

C. SELECTION OF SEED REARER

The rearing of seed cocoons requires technical skill. Seed Cocoons must be healthy, hygienic and preserve the racial characters. The seed rearer is to be selected on the following criteria.

The rearer should have a scientific knowledge of the Grainage operation and silkworm rearing. He should have an interest in sericulture and co-operate with Grainage personnel.

The rearing house should be located in an area suitable for pure breed rearing with optimum rearing conditions. It should be free from germs of silkworm diseases.

D. PROCUREMENT OF SEED

Seed cocoons are those which are produced under ideal climatic condition, free from diseases, exclusively for the purpose of reproduction, seed cocoons should conform to the racial characteristics. In practice these seed cocoons are raised in seed areas.

As the quality of seed cocoons determines the quality and productivity of seeds, It is very important that the cocoons procured for seed preparation should be of high standard. Keeping this in view, Standards are fixed in respect of different races and seasons.

THE IMPORTANT NORMS FOR PRODUCING THE SEED COCOONS ARE AS FOLLOWS:

1. Purchase cocoons which have been closely watched by the extensive staff and health certificate affixed on inspection card.
2. Gut examination of the pupae must be conducted before purchase.
3. The seed crop should be free from diseases especially pebrine.
4. Cocoons showing even a slight incidence of pebrine must be rejected.
5. The seed crop should have been reared under ideal conditions and fed with nutritious mulberry leaves.
6. The seed crop should have a good survival rate.
7. The seed crop should have a high pupation rate, but do not purchase cocoons which are with very heavy pupal weight that might lead to melting.
8. The cocoons should have good cocoon weight.
9. Cocoons which are not confirming to the characters of the race should not be purchased.
10. Crops showing an average yield of cocoons and above as fixed by the norms should only be purchased.
11. Rates are fixed as per the standards.
12. Purchase officer must certify for the quality of the cocoons and its disease freeness.
13. Details of rearer, quality and quantity of cocoons, race, spinning date, cost, and total amount paid, and name of the market are recorded and sent to the Grainage along with cocoons.

TRANSPORTATION OF SEED COCOONS:

It means carrying seed cocoon safely from producing centers to egg processing centers. Safe transportation is necessary not to affect pupae and cocoons which hamper the moth emergence. The seed cocoons are harvested on the 5th or 6th day after spinning.

This stage is suitable for transportation. If the pupa has turned dark brown in colour and if it is hard to touch then the seed cocoons are fit for transportation. Transportation at the late pupal stage causes damage.

The best time for transportation of seed cocoons is the cooler hours of the day i.e., early morning or late evening. Hot days damage the pupae due to the heat. Such cocoons emerge weak moths which either die or lay poor eggs. Laying and hatching of such eggs will be irregular.

Seed cocoons should be loosely packed in the containers (cloth bags, Bamboo conical basket, and plastic perforated bins) So as to allow sufficient space for aeration. Each container is perforated and the containers are placed horizontally one upon the other in rows.

SELECTION OF SEED COCOONS:

As soon as the cocoons arrive at the grainage, they are checked for their quality and quantity as per the details received from the cocoon market. Thus a preliminary examination is conducted to make sure that the seed cocoons are free from pebrine. From each batch of cocoons from the rearer, about 20 pupae are collected and gut examination is conducted for pebrine disease.



- **SEX SEPARATION.**

For preparation of hybrid eggs, the male and female moths of two different races are crossed (bi-voltine, multi-voltine races). When the moths emerge, the male and female moths of the same race copulate, this produces eggs of a pure race. To prevent this, it is necessary to separate the sexes before emergence so males and females are kept in separate trays. The sex separation can be done at larval stage, pupal stage and moth stage.

- **LARVAL STAGE**

In larval stage the abdominal segments carry the sexual markings, which develop in the 4th and 5th instars, in the 8th and 9th segments on the ventral side of the larva.

In Female larva a pair of milky white spots present in 8th and 9th abdominal segments (Ishiwata's glands). In the male a small milky white spot (Herold's gland) appear at the centre of the ventral side between the 8th and 9th abdominal segments.

- **PUPAL STAGE**

When compared to larval stage it is easy to determine the sex in the pupa stage. Female pupae are larger in size with a broad abdomen and an 'x' mark can be seen on the ventral side of the 8th abdominal segment.

Whereas the males are smaller in size having a narrow pointed abdomen with a small dot like mark on the ventral side near the top demarcation line of the 9th abdominal segment.

- **MOTH STAGE**

Female moths are inactive, larger in size and possess narrow antennae without bristles. The abdomen is bloated due to the presence of eggs. At the posterior end of the abdomen there is a protractile knob like ovipositor.

Male moths are distinct by their smaller size, narrow abdomen and broader antennae with bristles known as Pectinate antennae. Male moths are very active and are seen fluttering their wings in search of females. At the posterior end of the abdomen there is a pair of hooks like structures called Harps or Claspers.

- **COUPLING / MATING / PAIRING:**

After emergence of moths, the males and females are picked and allowed to mate as per the desired combinations. This is known as Pairing.

The female moths are collected in the tray into which the desired male moths are left over the female gently. The moths pair within 5-10 minutes. The left over male moths are collected into another tray. If the males available in excess, they should be preserved at 5°C for later use.

Each paired moths are kept in cellules arranged in a tray. All the trays are kept in dark room for about 3-4 hrs with a temperature of 25°C and 75-80 % humidity.



GRAINAGE REGISTERS/RECORDS:

It is essential that accurate data of seed production are maintained in the Grainage. This will help in efficient management, control of quality.

The following are the types of records to be maintained in Grainage.

A. SEED COCOON REGISTER

The register should contain information about the parental race, yield, silk and its quality details, bad cocoon percentage, seed rearer details, details of seed cocoon seller, cocoon purchase rate, weight of cocoons purchased and their cocoon quality details.

B. MOTH EMERGENCE REGISTER

The register should contain details of moth emergence such as number of moths obtained each day, number of moths kept for pairing, percentage of emergence, number of moths left for egg laying under each combination are recorded. This will help in forecasting the production and planning for hibernation. These data are to be maintained separately for different batches.

C. MOTH EXAMINATION REGISTER

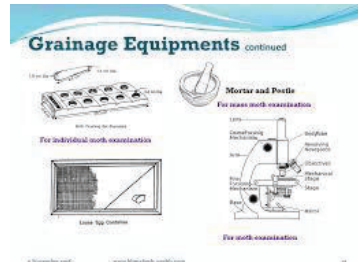
Maintenance of this record is important, as it provides information on the number of moths examined, type of examination conducted, presence of Fibrin, if present, percentage of infection, whether the batch is to be rejected.

D. EGG PRODUCTION REGISTER

1. Information pertaining to the quality and quantity of eggs laid, unfertilized eggs rejected, quality fit for distribution, type of egg (hibernated and no hibernated) are maintained.
2. E. Hibernation and Refrigeration Register
3. In this register contains details of refrigeration, hibernation, date of release, temperature in cold storage and number of moths kept in refrigerator.



GRAINAGE EQUIPMENTS



The Grainage equipments are designed for a specific function. These equipments are made in such a way to transport and handle easily. Equipment is used to control disease incidence, to sterilize, to check the attack of ants, to check light, to keep moths undisturbed. The Grainage equipment and their uses are as follows.

1.COCOON PRESERVATION STAND

Cocoon preservation stand are made of wood or bamboo or iron. It is easy to move them from place to place. This rack measures 228.6 cm height, 144.8 cm length and 61 cm breadth and should have 10-12 shelves with a space of 20 cm between each self. The seed cocoon preservation trays are arranged on the shelves and each stand can accommodate 10 trays. It is used for preservation of seed cocoons on the shelves.

2. COCOON PRESERVATION TRAYS

Grainage trays are portable receptacles for keeping seed cocoons, paired moths and egg laying moths.

TRAYS ARE OF TWO TYPES:

- (a) Round tray
- (b) Rectangular tray

a. ROUND TRAY:

It is made of bamboo and easy to handle. It is used for preservation of seed cocoons. Each tray is of 137.2 diameter in size, depth 6.5 cm

b. RECTANGULAR TRAYS OR WOODEN TRAYS:

It is made of wood, ply wood or wire mesh bottom. It is used in Grainage for pairing and ovi-position of moths. It measures 91.5cm length and 61 cm breadth.

3. ANT WELL:

It is made of concrete or stone blocks, 21 cm square and 8 cm high with a deep groove of 4 cm running all round the top. The leg of Grainage stand rest on the centre of the block and water is poured into the groove to stop 228.6 Cm (Height) 144.8 cm 7.6 cm x 5.1 cm 2.5 cm x 9.1 cm 6.4 cm (Height).

Antwell are used to control the attack of ants. Ants cause lot of damage to cocoons and moths.

A. Cocoon Preservation Tray Round

B. Wooden Tray

C. Grainage - Wiremesh Tray 2.5 cm (Thick) 50.8 cm 8.91 61.0 crawling of ants on to cocoon preservation trays. Each stand leg must rest in a well.

4. CELLULE:

It is made of plastic which is black in colour measuring 3.2 cm in height and 5.1 cm in diameter. The copulate females are kept on an egg sheet and are covered with a cellule. It provides semi-dark condition around the paired moths and ovi-positing female moths. It increases the fertility and egg laying capacity.

5. MOTH CRUSHING SET:

It is made of porcelain, having 10 mortars and 10 pestles. It is used for crushing the larva, pupa and moths to prepare smears. The supernatant liquid after centrifuge is examined under the microscope for identification of pebrine spores. Diseased moth eggs are discarded before processing. The same work can also be done by moth crushing machine and pebrine separator.

6. MICROSCOPE:

Compound microscope is used for conducting mother moth examination after ovi-position and pupal examination to detect pebrine spores. This microscope is having 600 magnifications.

7. WET AND DRY BULB THERMOMETER:

It is used to measure the room temperature and relative humidity in the Grainage.

DIAL HYGROMETER:

It is used to measure room humidity directly

8. WASH BASIN OR BASIN STAND:

It is a tripod stand with a height of 86.5 cm to hold a basin (30.5 cm dia). The basin is filled with 2 % formalin. This liquid is to wash and sterilize hands while entering into the room. It is kept nearer to cocoon, moth preservation room and egg laying room.

9. ACID TREATMENT BATH:

This equipment is used for treating the uni-voltine and bi-voltine layings to break hibernation and enables them to hatch as usual.

10. CRATES:

Crates are used for preserving the male moths after emergence.

11. REFRIGERATOR:

Refrigerator is used for synchronization of moths and also to preserve male moths which can be used for second pairing.

12. INCUBATOR:

Incubator are used to incubate the silkworm eggs at 23-25°C temperature and 80-85% relative humidity for uniform development of the embryo.

13. SPRAYER:

It is used for disinfecting the Grainage building and equipments.

14. MOUTH MASK:

The silkworms have plenty of scales on body and wings. These scales are spread in the moth emergence room during their emergence. Thus mask is used to prevent inhaling of scales, dust and formalin fumes by workers.

15. DEFLOSSING MACHINE:

Deflossing machine is used to defloss the cocoon before moth emergence. It is operated by pressing the pedal. The deflossed cocoons are collected and preserved.

16. COCOON CUTTING MACHINE (CCM):

It is used to cut the cocoon and separate the pupae which is one of the important grainage activity. It is mechanized equipment .

17. FORMALIN MAT (FOOT CLEANING TRAY):

A gunny bag or cloth piece is spread in a iron sheet tray having 2 % formalin or any other disinfectant and kept in front of the door. While entering into the room one should keep the foot in the tray for disinfection.

18. MOTH TESTING TABLE AND STOOL:

These are used to keep microscope during moth examination. The table is to keep moth crushing set, slide box, cover glass packet, water or KOH and observation note book. The stool is used for sitting during moth examination.

19. OTHER EQUIPMENTS:

Other equipments like air cooler, slide box, cover glasses, electric heater, egg sheets, loose egg boxes, hydrometer, clock are also required for Grainage.

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